GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH

LYSA LIEW WAN JIN

FACULTY OF INFORMATION SCIENCE & TECHNOLOGY

MULTIMEDIA UNIVERSITY

JULY 2025

GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH

BY

LYSA LIEW WAN JIN

THE PROJECT REPORT IS PREPARED FOR

FACULTY OF INFORMATION SCIENCE & TECHNOLOGY
MULTIMEDIA UNIVERSITY
IN PARTIAL FULFILLMENT
FOR

BACHELOR OF COMPUTER SCIENCE
B.C.S. (HONS) ARTIFICIAL INTELLIGENCE

FACULTY OF INFORMATION SCIENCE & TECHNOLOGY

MULTIMEDIA UNIVERSITY

JULY 2025

© 2025 Universiti Telekom Sdn. Bhd. ALL RIGHTS RESERVED

Copyright of this report belongs to Universiti Telekom Sdn. Bhd as qualified by Regulation 7.2 (c) of the Multimedia University Intellectual Property and Commercialization policy. No part of this publication may be reproduced, stored in or introduced into a retrieval system, or transmitted in any form or by any means (electronic, mechanical, photocopying, recording, or otherwise), or for any purpose, without the express written permission of Universiti Telekom Sdn. Bhd. Due acknowledgement shall always be made of the use of any material contained in, or derived from, this report.

DECLARATION

I hereby declare that the work have been done by myself and no portion of the work contained in this thesis has been submitted in support of any application for any other degree or qualification of this or any other university or institute of learning.

_____

Lysa Liew Wan Jin

Faculty of Information Science & Technology
Multimedia University

Submission Date: 12: 06: 2025

Submission Time: 11: 00

ACKNOWLEDGEMENT

I would like to thank everyone who had contributed to the successful completion of this project. I would like to express my sincere gratitude to my supervisor, Miss Pa Pa Min for her outstanding leadership and constant guidance throughout the progress of this project. My understanding and approach have been developed thanks to her invaluable advice, guidance and knowledge. I sincerely appreciate her patience, depth of knowledge, time and effort she has put into my progress. Her support has enhanced my academic development and also motivated me to continuously strive for excellence.

I would also like to express my gratitude to my loving parents and friends who had helped and given me encouragement. They have taken their time and made the effort to make sure that I am motivated to do my work to ensure that it is up to satisfactory levels. This showed their commitment to my development and their sincere interest in it is greatly appreciated.

ABSTRACT

Gait recognition is quickly emerging as an innovative form of biometric technology due to its ability to function remotely and without the need for the individual to actively engage. This project research into demonstrating the application of deep learning algorithms that use silhouette representations as input. The main objective is to overcome the problems of traditional gait recognition techniques which is machine learning that has low accuracy. This project uses CASIA-B dataset, which is a multiview dataset that includes different walking conditions. The transformations of raw silhouettes into Gait Energy Image (GEI) which are two-dimensional representations of gait patterns are done. The primary outcome demonstrated the implementation of a pre-trained Vision Transformer (ViT) model for gait detection tasks as it showed that GEI works well in gathering spatiotemporal features. Several preprocessing steps has been done to ensure that the training data is consistent and robust. There are several experiments conducted under different walking conditions across the 90° viewing angle. The performance of model is evaluated using standard evaluation metrics and it proves that this model does not only achieve high accuracy in one walking condition, but across different combinations of conditions. The comparative analysis with existing gait recognition models further highlights the adaptability and suitability of this pre-trained ViT. Overall, this project creates an adaptable and effective system for recognising gait that represents a rapidly evolving kind of biometric technology through the deployment of deep learning model.

TABLE OF CONTENTS

DECLARATION ..... **III**

ACKNOWLEDGEMENT ..... **IV**

ABSTRACT ..... **V**

TABLE OF CONTENTS ..... **VI**

LIST OF TABLES ..... **VIII**

LIST OF FIGURES ..... **IX**

LIST OF ABBREVIATIONS/ SYMBOLS ..... **X**

LIST OF APPENDICES ..... **XI**

CHAPTER 1 INTRODUCTION **1**

1.1 Overview 1

1.2 Problem Statement 2

1.3 Project Objectives 3

1.4 Project Scope 4

1.5 Report Organisation 5

CHAPTER 2 LITERATURE REVIEW **6**

2.1 Gait Recognition: An Overview 6

2.2 Gait Recognition Approach 6

2.3 Gait Recognition Representation 8

2.4 Traditional Gait Recognition Methods 9

2.5 Deep Learning for Gait Recognition 12

2.6 CNN Architectures in Gait Recognition 16

2.7 Datasets for Gait Recognition 19

2.8 Performance Evaluation Metrics 20

2.9 Summary 23

CHAPTER 3 METHODOLOGY **25**

3.1 Overview 25

3.2 Research Framework 25

3.3 Software and Hardware Setup 36

3.4 Project Timeline 37
3.5 Summary 38

CHAPTER 4 EXPERIMENTAL DESIGN AND ANALYSIS 40

4.1 Overview 40
4.2 Dataset 40
4.3 Data Preprocessing 41
4.4 Model Training 44
4.5 Hyperparameter Tuning 45
4.6 Model Validation 46
4.7 Experiments 47
4.8 Comparison with Existing Methods 67
4.9 Summary 69

CHAPTER 5 CONCLUSION 71

5.1 Conclusion 71
5.2 Future work 72

REFERENCES 73

APPENDICES 76

LIST OF TABLES

Table 2.1: Advantages and Disadvantages of Traditional Gait Recognition Method 12
Table 2.2: Advantages and Disadvantages of Deep Learning Recognition Method 16
Table 2.3: Confusion Matrix Format 22
Table 2.4: Summary of Gait Recognition Based Studies 23
Table 3.1: Representation of Component 27
Table 4.1: Hyperparameters 46
Table 4.2: Experiment 1 Results 48
Table 4.3: Normal walking Results 52
Table 4.4: Wearing a coat Results 56
Table 4.5: Carrying a bag Results 60
Table 4.6: Experiment 3 Results 64
Table 4.7: Comparison between existing models and our pre-trained ViT 68

LIST OF FIGURES

Figure 2.1: CNN architecture 13
Figure 2.2: GAN architecture 13
Figure 2.3: Autoencoders architecture 13
Figure 2.4: GaitSlice architecture 16
Figure 2.5: CNN architecture 17
Figure 2.6: Deep CNN architecture 18
Figure 2.7: Efficientnet-b0 architecture 19
Figure 3.1: Gait Recognition Framework 26
Figure 3.2: GEI for all three conditions 28
Figure 3.3: Pre-trained ViT-GR Architecture 29
Figure 3.4: Transformer Encoder 31
Figure 3.5: Gantt Chart 38
Figure 3.6: Gantt Chart 38
Figure 4.1: First individual with normal walking condition 42
Figure 4.2: 62nd individual with carrying a bag condition 42
Figure 4.3: 124th individual with wearing a coat condition 43
Figure 4.4: Training and Validation Loss 49
Figure 4.5: Training and Validation Accuracy 50
Figure 4.6: Confusion Matrix 51
Figure 4.7: Training and Validation Loss 53
Figure 4.8: Training and Validation Accuracy 54
Figure 4.9: Confusion Matrix 55
Figure 4.10: Training and Validation Loss 57
Figure 4.11: Training and Validation Accuracy 58
Figure 4.12: Confusion Matrix 59
Figure 4.13: Training and Validation Loss 61
Figure 4.14: Training and Validation Accuracy 62
Figure 4.15: Confusion Matrix 63
Figure 4.16: Training and Validation Loss 65
Figure 4.17: Training and Validation Accuracy 66
Figure 4.18: Confusion Matrix 67

LIST OF ABBREVIATIONS/ SYMBOLS

ANN	Artificial Neural Network
CCA	Canonical Correlation Analysis
CCR	Correct Classification Rates
CGI	Chrono-Gait Image
CMC	Cumulative Match Characteristics
CNN	Convolutional Neural Network
DBN	Deep Belief Networks
EER	Equal Error Rate
FAR	False Acceptance Rate
FDEI	Frame-Difference Energy Images
FFN	Feed-Forward Network
FN	False Negative
FP	False Positive
FRR	False Rejection Rate
GAN	Generative Adversarial Networks
GEI	Gait Energy Image
GEnI	Gait Entropy Images
GMM	Gaussian Mixture Models
LDA	Linear Discriminant Analysis
LSTM	Long Short-Term Memory
MHSA	Multi-Head Self-Attention
OaA-SVM	One-against-all-SVM
PCA	Principal Component Analysis
PEI	Period Energy Images
RBF	Radial Basis Function
RNN	Recurrent Neural Networks
SGEI	Skeleton Gait Energy Image
SVM	Support Vector Machines
TN	True Negative
TP	True Positive
ViT	Vision Transformer

LIST OF APPENDICES

Appendix A: FYP Meeting Log 76
Appendix B: Technical Paper 82
Appendix C: Checklist for FYP Final Report Submission 83

CHAPTER 1

INTRODUCTION

1.1 Overview

One of the foundations of security is the ability to be recognised. A person may be identified using a variety of biometrics methods. According to Kececi, Yildirak, Ozyazici, Gulsen, Ayluctarhan and Agbulut (2020), this method helps to automatically identify a person by distinguishing between individuals through the use of identification technology. Biometric characteristics of humans can be classified as either physiological or behavioural characteristics. Asif, Tiwana, Khan, Ahmad, Qureshi and Iqbal (2022) presented some examples of physiological features include fingerprints, ears, face, iris, and passwords whereas behavioural features are voice, signature, and gait. The behavioural features may be captured invisibly and non-invasively using a low-resolution camera system as they rarely require human contact (Asif et al., 2022). This is because some biometrics rely on human interaction to gather data and it can be bothersome.

Some users may feel upset as their identity is being verified quickly in such a scenario. However, falsifying biometric data is simple and this led to a surge in research towards more hidden and safer biometrics. Zou, Wang, Y., Wang, Q., Zhao and Li (2020) discovered that a biometric system needs to be able integrate in as little as possible in order to perform distinct functions like recognising a person. Gait recognition is a method of identifying an individual based on their physical and personality traits just by observing how they walk (Kececi et al., 2020). Each gait has its own set of characteristics where it identifies an individual without their assistance or physical contact with the system. This is a unique feature due to its unobtrusive design. The gait biometric method which involves figuring out individuals from their gait pattern has been widely acknowledged and used. Khan, Farid and Grzegorzek (2021) stated that the capacity to function from a distance and effective in lower-resolution films makes it more reliable.

1.2 Problem Statement

Since the importance of silhouette images is not entirely understood, it is hard to determine how they contribute to improve the effectiveness and dependability of gait detection systems in comparison to regular photos. The general and possible practical applications using silhouette can improve its development in real-world settings through the understanding of relevance of the entire technology (Song , Yun and Li, 2024).

There is a lack of advancement and limitation in the features found in traditional gait recognition systems (Parashar, A., Parashar, A., Ding, Shabaz and Rida, 2023). When encountered with several situations, traditional methods particularly unable to reach high accuracy and flexibility as they are basic. Traditional models are also inefficient when dealing with huge datasets and perform poorly in dealing with complicated real-world situations due to lack of scalability and accuracy.

The effectiveness, reliability, and interpretability of some gait recognition models are uncertain as some were created without conducting thorough evaluations (Parashar et al., 2023). How effectively these models adjust in real-world settings and how well they function under different conditions cannot be properly evaluated without performance measures. Therefore, the needs of evaluation in order to develop gait detection technologies that meet real-world requirements are highly important.

1.3 Project Objectives

The following objectives present an overall structure for the current project to address the problem stated. This can help provide information on how to enhance the proposed system even further. These are the primary objectives of the project:

- To study existing research on gait recognition with a focus on methods that make use of silhouette images.
- To develop a deep learning model that can identify individuals by their different gait patterns from silhouette images.
- To assess the performance and accuracy of developed model in recognising gait patterns.

This is because silhouette-based methods maintain the important features for evaluation of gait while understanding the different external patterns of individuals. Other than that, the importance of gait recognition such as advantages, disadvantages, and potential uses in the real world are also analysed.

The system is tested on a multiview dataset that contain a wide range of situations to determine its resilience and flexibility. Important performance measures will be examined to gain a thorough knowledge of the model's capabilities and problems.

1.4 Project Scope

This project aims to review existing studies on silhouette-based gait identification techniques since the understanding of silhouette images are not fully developed. Moreover, a deep learning model that can recognise people based on their different gait structures will be developed to solve the issue related to insufficient capabilities of traditional systems. Additionally, the effectiveness of model in different environments are assessed to evaluate the practical performance. The primary goal is to create a gait recognition system that is reliable and efficient by combining ViT with GEI. The CASIA-B, which is a well-known multiview dataset is used in this project. It ensures thorough evaluation of the model using distinct gait sequences obtained under different situations, including various points of view, type of clothing, and carrying a bag scenario. The main input data for ViT model is GEI, which average silhouette pictures throughout a gait cycle to combine spatial and temporal gait patterns into one frame. In order to preserve spatial information, the input GEI images are separated into patches, placed into spaces with more dimensions, and then enhanced using positional embeddings. The transformer encoder consists of MHSA, FFN and activation functions to analyse these patches in order to extract useful gait data. Layer normalisation and residual connections provide consistent training and better gradient as further improvements. In order to accurately identify individuals, the classification head uses the retrieved characteristics to provide probabilistic outcomes. Finally, the effectiveness of the model in terms of reliability and dependability are evaluated using metrics to measure how well is the performance.

1.5 Report Organisation

The report is organised into five chapters that address different topics of the study. This allows for a thorough summary of the research for gait recognition.

Chapter 1 of the report provides an overview of the project, including the introduction, problem statement, objectives, project scope and report organisation. This chapter is the foundation for understanding this gait recognition project.

Chapter 2 provides relevant literature by analysing previous studies on gait recognition, traditional machine learning models and deep learning algorithms. Architectures of CNN, datasets and metrics for measuring performance. The literature review sets the work in relation with previous research and highlights its importance.

Chapter 3 presents the methodology which describes the research framework and architecture used in this project. This includes the details of data acquisition, preprocessing and ViT model. The technique is presented in a neat and organised way to guarantee consistency.

Chapter 4 which is the experimental design that explores the dataset used and its components. The testing of experiments is conducted to observe the result achieved and evaluate the performance of models.

Chapter 5 is the conclusion which summarises the work completed to achieve the objectives of this project. The future work is also suggested to ensure that it will be performed to make the gait detection system even better.

CHAPTER 2

LITERATURE REVIEW

2.1 Gait Recognition: An Overview

A person's gait or the way they walk is an important biometric characteristic that may be identified as demonstrated by Narayan, Awasthi, Fautima, Faiz and Srivastava (2023). Gait recognition is the process of detecting individuals based on their walking characteristics. Individuals have varying gait cycles or walking cycles as discussed by Kumar, M., Singh, Kumar, R., Goel and Kumar, K. (2021). Saleem, Khan, Alhaisoni, Tariq, Armghan, Alenezi, Choi and Kadry (2021) discovered that gait identification is influenced by camera views, carrying load, lighting settings, clothing variety, walking rate, and shadows underfoot.

Içik and Ekenel (2021) stated that gait recognition can operate from a distance and do not need additional requirements from the cooperation of the individual as compared to other biometrics recognition. Zou et al. (2020) showed that other biometrics such as iris, fingerprint, face, and voice are intrusive to users because they require user cooperation to collect data. In these situations, a user may feel offended and quickly realise that their identification is being checked.

Içik and Ekenel (2021) found that an advantage of gait recognition is that someone else's gait cannot be mimicked precisely. Some tactics such as covering the face or wearing gloves can make a person unrecognisable. However, concealing one's walk is more challenging. This is because certain biometrics like face recognition are easily falsified and exploited using images or videos of the target face as researched by Zou et al. (2020).

2.2 Gait Recognition Approach

Khan, A., Khan, A. M., Javed, Alhaisoni, Tariq, Kadry, Choi and Nam (2021) demonstrated that there are two methods for recognising human gait which are model-

based and model-free. The model-based method uses previous data to guide movement of humans. This method examines human behaviour by looking at upper and lower body components and movement of joints. The model-free method creates skeletons or postural generation which are representations of the human body. However, it is easier to build and takes less computing time to use the model-free technique.

Saleem et al. (2021) researched that model-based and model-free methods are the two main subcategories of human gait recognition. Human gait is categorised using model-based methods such as movement and body shape models. A human body is described using an architectural model and define gait by tracking various body components and movement of joints over time using basic mathematical concepts including stick-figure, interconnected pendulum and elliptical fitting approaches. The model-free approach which is easy to use because it requires less computing power uses the gait time to determine the attributes of the object silhouette.

According to Kumar et al. (2021), there are two types of approaches which are model-free and model-based. Model-free method uses a binary silhouette to identify the gait of a person. A silhouette is a single-coloured, solid-shaped picture of someone or object. Typically, the inside of silhouette has fewer features and is coloured black. Therefore, applying background modelling methods is necessary for the extraction of silhouettes. Model-based approach uses information about a person's joints and other body elements. The quality of gait sequence needs to be high for this method. Therefore, a multi-camera arrangement is required to give the system high-quality gait sequences.

A. Khan et al. (2022) demonstrated that the current gait identification methods may be grouped into two methods which are model-based techniques and model-free techniques. Model-based methods use movement and structure of the body models to create a gait characteristic. This method is independent of scale and vision. The skeletal models explain how to represent a person's torso, upper limbs, and lower body parts by calculating various measures on these areas like width and height. The motion analysis model which is applied for gait detection, focusses on the movement data of the body of a human or its numerous parts throughout time. However, model-free

approaches do not involve any kind of human structure or geometrical knowledge. It uses the captured gait footage to directly extract motion trends in terms of form and spatial direction. An individual can be identified by using a template image created by combining binary silhouettes of human from gait video sequences over the gait cycle.

2.3 Gait Recognition Representation

2.3.1 Image-based

Içik and Ekenel (2021) discovered that the image-based gait is interpreting the actions or body form of an individual from the images and high video quality is not required. Obtaining silhouettes of human by eliminating the backgrounds from the frames is the most popular fundamental step in these methods. These human silhouettes are used to generate a gait pattern. The sequences of walking are often represented by a single image for image-based classification because walking is a continuous action. The GEI is a representation which can be calculated by just averaging every human silhouette across a gait cycle. The whole time between the first step a foot takes when touching the ground and the subsequent steps that follow is known as the gait cycle. GEI results in minimal computing cost and great accuracy but the drawback is the loss of certain temporal data. In general, image-based methods are advantageous because they can easily handle identical-view photos, which are acquired from the same angle, while other methods are impacted by angle variations.

Chao, Wang, He, Zhang and Feng (2022) researched that gait can be seen as a single picture by compressing all gait silhouettes into a single image known as gait template for the purpose of gait recognition. The compression technique leaves out important characteristics like temporal information and fine-grained spatial information even when different existing gait templates encode information as efficiently as practicable. Compressing gait data into a single image such as a CGI or a GEI is the aim of template development. These methods first evaluate the human silhouettes in each picture by getting rid of the background in order to create a template. The template matching process extracts the gait representations from a template image.

Sepas-Moghaddam and Etemad (2023) discovered that templates average the silhouettes across a minimum of one cycle of gait to collect time-based walking data across several silhouettes on a single map. The first network layer are also referred to as temporal templates can collect gait silhouettes. There are several temporal templates such as GEI, CGI, FDEI, PEI, and GEnI. GEI takes the average of a period or succession of gait patterns. The contour of each gait picture is extracted using CGI which are subsequently encoded with a multi-channel mapping algorithm to create a single map. FDEI which use denoising and grouping techniques to maintain the kinetic information especially in cases where the silhouettes are not occupied. PEI are a generalisation of GEI that preserve more of both temporal and spatial information by utilising a multi-channel mapping function based on the amplitudes of frames and GEnI compute entropy for every pixel in the gait frames and averages them into a single gait template.

2.3.2 Video-based

Kumar et al. (2021) demonstrated that data from videos contains the full motion and shape of an individual's silhouette image. Silhouette is regarded as the main approach for this representation of video data characteristics as it converts the image by obtaining each pixel of an element and displaying its features in silhouette. Typically, the quality of the video determines the scale and perspective. A person's structural stride characteristics have been used as representations of the human body. There are two parameters such as step and stride length. Steps are measured in minutes while stride length is calculated in meters. These parameters are taken from the video of the gait to acknowledge the individual.

2.4 Traditional Gait Recognition Methods

Parashar et al. (2023) researched that machine learning is the traditional gait recognition methods as it requires manual feature engineering. Feature extraction is an important part of this procedure as it involves selecting relevant gait data information to distinguish between individuals. There are various common gait detection algorithms as discovered by Parashar et al. (2023) that are based on machine learning approaches that involve manual feature engineering work. An algorithm such as

Histogram of Orientated Gradients (HOG) which extracts features defining the shape and properties of objects in photos. It can accurately show the shape of the human body as well as leg movements. Next, PCA is a technique for projection of data onto a lower-dimensional space in order to reduce dimensionality while maintaining important information. Lastly, Dynamic Time Warping (DTW) examines the similarity of time-series data and compare the patterns of gait between individuals.

Kececi et al. (2020) demonstrated that machine learning automates both the framework and the operation of algorithms to make predictions using the data provided as inputs. These algorithms operate by creating a model that makes data-driven decisions using data from a small sample rather than relying solely on static computer instructions. Kececi et al. (2020) stated that ten machine learning algorithms are used in their research and only three of these algorithms were successful. The machine learning algorithms stated by Kececi et al. consist of RIPPER which is most common rule-based learning algorithms. ANNs which is organised in layers is known as multilayer perceptron. Next, decision tree is a framework that can split a large data into smaller sets using a set of decision rules. Bootstrap aggregation or bagging is a meta method used in the field of machine learning to increase stability and accuracy. Next, one classifier uses a linear regression method for categorisation. The random tree operator is identical to the operator in decision tree with single change which is only a random subset of attributes is provided for each split. Naïve Bayes classification uses probability-based algorithms to classify presented data. According to Kececi et al. (2020), the three successful algorithms are random forest which attempts to improve classification accuracy by combining many decision trees. Furthermore, IBI which employs the standardised Euclidean distance to discover the sample that is closest to the requested sample and assigns it to the same class. Lastly, BayesNet which uses a directed acyclic graph as its graphical model structure.

Chao et al. (2022) discovered that machine learning approaches helps to extract gait representation from template images during the template matching procedure. Template generation procedure combines gait data into one image such as a GEI or a CGI. CCA and LDA were used as machine learning approaches. Euclidean distance or other metrics learning techniques are used to determine how similar two

representations are to one another. In order to get a view-invariant representation, the templates are projected into a latent space using the proposed view-invariant discriminative projection (ViDP). This is done after the view transformation model (VTM) has learnt how to project between different perspectives. Lastly, they use a classifier such as a SVM or nearest neighbour classifier to label the template based on the calculated distance.

Nazari, Mohajer, Nahavandi and Khosravi (2022) researched that machine learning-based systems used manually derived features to identify the gait period from signals that are input. These methods include LDA, SVM, GMM, and Decision Trees. PCA is an effective method for decreasing the input data dimensions which can save computing expenses for applications running in real time. Nazari et al. (2022) discovered that Gaussian Naive Bayes, Decision Trees, Random Forests, and LDA models were train with handcrafted features. Model hyperparameters were tuned for every experiment using a training dataset and random search approach. The ideal model was recorded based on the test dataset and performance time. Models were developed using the same machine learning methods and reduced data from PCA. In their research, the models have been trained on several number of principal components.

Asif et al. (2022) presented SVM algorithm to categorise various photos according to view angle in various scenarios. Features have been extracted by doing research of covariate variables. This approach consists of many primary stages including pre-processing, feature extraction, training, validation, and labelling. GEIs are the inputs and they have been pre-processed so that random sampling can be done without changing the images. The histogram of orientated gradients is then used to extract features. The training step involves training SVM classifiers to recognise the input as this helps to improve the performance throughout the evaluation phase. During the testing stage, a SVM model was used for various classifications of the variables. This approach works well when it comes to functionality for recognising human gait.

Table 2.1: Advantages and Disadvantages of Traditional Gait Recognition Method

Advantages	Disadvantages
Perform well with less data	Need manual feature engineering
Require less computer resources	Time-consuming
More interpretable	May not gather all important data
Suitable for applications that require interpretability	Lacks the ability to detect complicated patterns in data
	Restricts performance

2.5 Deep Learning for Gait Recognition

Filipi Gonçalves dos Santos, de Souza Oliveira, Passos, Gonçalves Pires, Silva Santos, Pascotti Valem, Moreira, Santana, Roder, Papa and Colombo (2021) researched that deep learning is a cutting-edge machine learning technique that excels at surpassing previous benchmarks. This new framework eliminates the need for manual feature extraction from researchers and achieves better gait recognition results. This helped to overcome the existing challenges and establishing way for future study. Filipi Gonçalves Dos Santos et al. (2021) proposed several deep learning architectures such as CNN, RNN, Capsule Networks, GAN, DBN, and Autoencoder for recognition of individual's gait. CNN are based on the visual brain and use convolutional neural networks to conduct operations on input data. Figure 2.1 shows the architecture of CNN. RNN compute neurone activity based on input of data and output of other neurones. Capsule Networks provides the use of hierarchy as it is a two-layer architecture which are convolutional encoder and fully connected linear decoder. GAN are made up of two separate networks which are the discriminator and the generator. Figure 2.2 shows the GAN architecture. DBN are stochastic neural network structures that are effectively suited for tasks that are generative. Generative neural networks called autoencoders include both encoders and decoders. Figure 2.3 shows the architecture of Autoencoders.

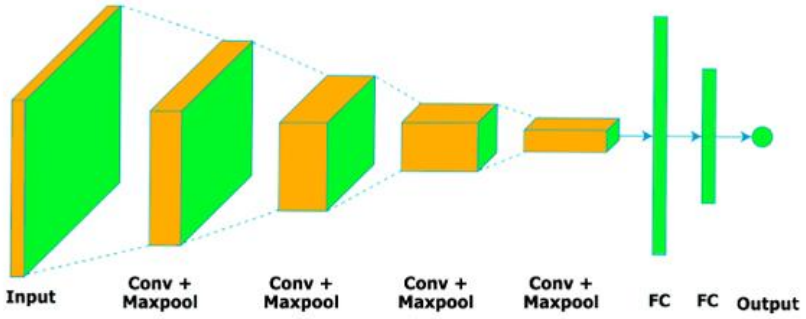


Figure 2.1: CNN architecture (Filipi Gonçalves Dos Santos et al., 2021)

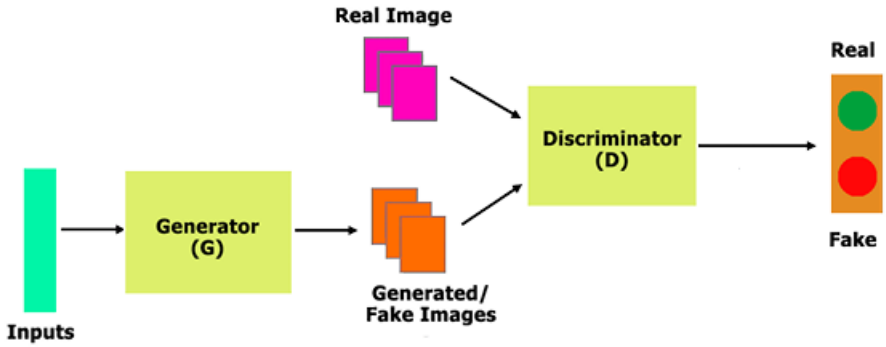


Figure 2.2: GAN architecture (Filipi Gonçalves Dos Santos et al., 2021)

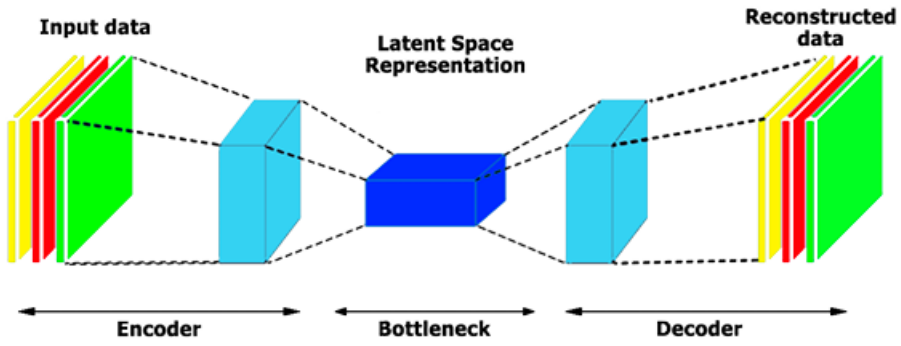


Figure 2.3: Autoencoders architecture (Filipi Gonçalves Dos Santos et al., 2021)

Narayan et al. (2023) researched a list of popular deep learning algorithms for gait recognition such as CNN, Capsule Networks, RNN, Autoencoder, DBN, and GAN. CNN are neural network that can recognise and identify between multiple features in an image by first assigning a priority to each feature. Capsule Networks is an advanced model of CNN that uses the routing-by-agreement technique in which only features that fit within its particular type are sent to the higher layer. RNN uses a hidden layer

as a portion of the previous sequence's information has been kept in this layer. Autoencoders are designed specifically to learn ways to represent higher-dimension input in lower dimensions. DBN consist of many layers of successively connected restricted Boltzmann machines. The output of restricted Boltzmann machines serves as the following input and each of it is trained until convergence. GAN implement an innovative approach of generative model development by using the generator model and discriminator model to present the task as a supervised problem.

Sepas-Moghaddam and Etemad (2023) discovered multiple deep learning architectures such as CNN, DBN, RNN, Deep Autoencoder, GAN, Capsule Networks, 3D CNN, Graph Convolutional Networks, and Hybrid Networks. CNN encode the body form shown as a silhouette or skeleton in a feature space. A DBN is a probabilistic generative neural network that extracts hierarchy-based representations from data sets used for training. RNN usually consist of a layer with several cells in which each representing an input element in the ordered sequence. Deep auto-encoder networks extract bottlenecks characteristics or hidden space structures through an encoder and decoder structure. GAN consist of a generator and a discriminator. The generator creates falsified samples to trick the discriminator. Capsule Networks are made up of capsules which are collections of neurones that encode the inbuilt view-point-invariant connections present in various sections of an object. 3D CNNs extract greater strength aspects that can withstand changes in camera angles and target appearance. Graph convolutional networks use dynamically organised graphs and convolution filters to expand CNNs to higher dimensions. As for hybrid networks, it is made up of multiple types of networks.

M. H. Khan et al. (2021) proposed that the basic principle of deep learning models is the use of a neural network that resembles how the human brain processes data which is made up of several neural connections like input, output, and hidden layers. There are three deep learning models which are CNN, RBF, and ANN. CNN were first created to address issues in computer vision and image processing. Convolution and pooling are the two fundamental operations. The feature map is extracted from the provided data by the method known as convolution whereas subsampling is done using pooling. RBF is a feed-forward neural network

with three layers. The input layer is the first layer that takes in network input followed by the hidden layer with an RBF non-linear activation function which is often implemented as a gaussian function and a certain number of nodes. The final layer is in charge of producing the network's output. An example of ANN is non-linear deep neural network built on auto-encoders to address the view-invariant gait recognition issue. The forward propagation over the trained network which is trained using spatiotemporal features of human walking produces the view-invariant gait properties of a validation occurrence.

Kukreja, Kumar and Kaur (2021) researched that a large number of deep learning models are capable of automatically classifying gait events and extracting characteristics. In this research, there are two deep learning models which are ANN and CNN. ANN uses feed-forward neural networks and is pattern-based. The three layers that make up this system are input, secret and output. The secret layer processes the input once it has been received by the layer of input. The output layer then generates the outcome. CNN is designed to meet the reduced criteria for processing and uses a device similar to a multilayer perceptron. This model is composed of three layers such as input, output, and hidden where each has several convolution, pooling, and fully connected layers. Video and image identification are the two applications for CNN.

Parashar et al. (2023) researched that the use of methods based on deep learning which allow for the automated learning of unique characteristics has improved gait recognition. There are three deep learning models which are CNN, RNN, and Autoencoder. CNN are widely used for recognition of images applications. It is capable of autonomously extracting information from gait photos for the purpose of gait recognition. RNN are commonly used for the analysis of data in sequence. The time sequence of information about gait can be analysed. Autoencoders are capable of learning without supervision as it learns a low-dimensional representation of gait data for categorisation in gait identification.

Table 2.2: Advantages and Disadvantages of Deep Learning Recognition Method

Advantages	Disadvantages
Performs well with large datasets	Require high computational resources
Extract features automatically	Difficult to interpret models
Can capture complex patterns	Require huge amount of data

2.6 CNN Architectures in Gait Recognition

Li, Qiu, Zhao,Zhan, Chen, Wei and Huang (2022) presented a model known as GaitSlice which uses interconnected slice features and parallel Residual Frame Attention Mechanism (RFAM) to flexibly choose the most important frames of every component of the body. This model has a new component called the Spatio-Temporal Slice Refinery (STSR) in addition to two traditional components called the Spatial Feature Extractor (SFE) and Spatial Horizontal Pooling (SHP). The Spatial Feature Extractor (SFE) is a CNN that comes with three blocks where each has multiple 2D layers of convolution, focal convolution layers, and pooling layers. Spatial Horizontal Pooling (SHP) uses the height dimensions to map features spatially into horizontal stripes, which are then combined into component features. Spatio-Temporal Slice Refinery (STSR) enhances the spatial and temporal properties of the components by obtaining the key frames from the relevant parts of the body. Figure 2.4 shows the architecture of the proposed model known as GaitSlice.

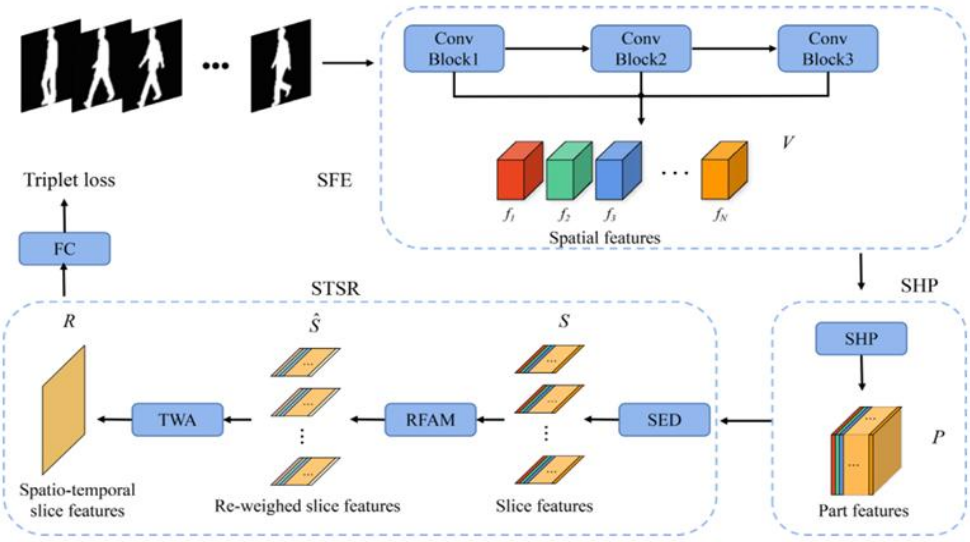


Figure 2.4: GaitSlice architecture (H. Li et al., 2022)

Sayeed, Min and Bari (2022) proposed that patterns of gait can be obtained using a CNN. To carry out feature training and classification, the CNN will process the input data which is GEI. GEI is the spatiotemporal gait representation that compresses gait characteristics into a single image. In order to maintain the movement of humans, GEIs transformed the series of gait silhouettes into a two-dimensional representation. Convolutional layers, pooling layers, hidden normalisation layers, layers that are fully connected, input, and output layers and other layers form a CNN. The most crucial layer is the convolutional layer which carries out operations involving convolution to identify the input image's feature patterns. The Softmax layer handles categorisation whereas pooling layers reduce the sample size of the input and fully connected layer's dimensions. Figure 2.5 shows the architecture of the proposed CNN model.

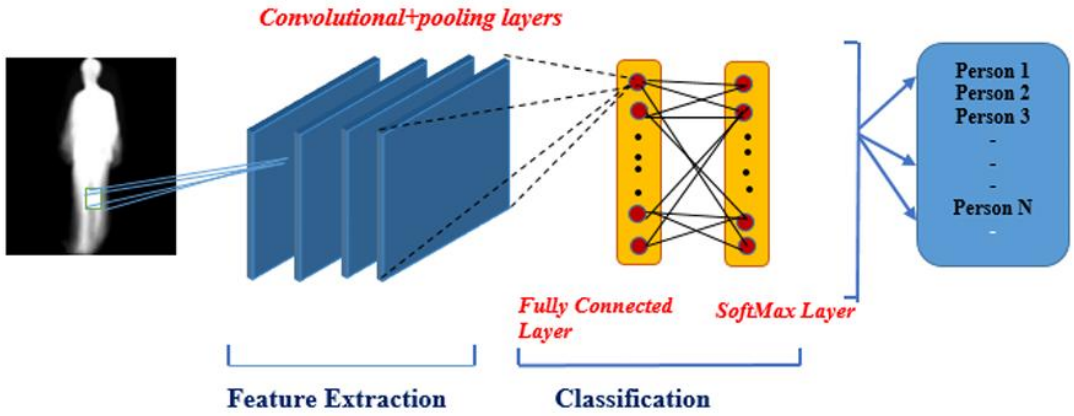


Figure 2.5: CNN architecture (Sayeed et al., 2022)

Nazari et al. (2022) demonstrated that Deep CNN can help to improve performance as compared to the traditional method. This is because Deep CNN extracts the important features instead of manually determining characteristics from the processed and downsampled data. Each EMG channel's 80 ms segments were subjected to feature extraction by two levels of the 1D convolutional layer. Every single layer was followed by a maxpooling layer that selected the most essential characteristics. To identify the basic trends in each of the signals in the five channels, the findings were put into two completely linked layers. To avoid overfitting, a 30% heavy ejection was utilised after each layer. In order to estimate the gait phase, the

output layer likewise has two neurones and is dense. The architecture of the Deep CNN model is shown in Figure 2.6.

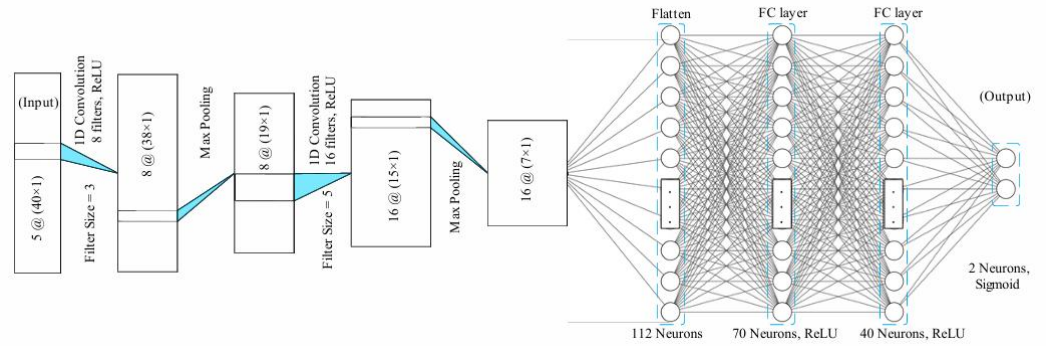


Figure 2.6: Deep CNN architecture (Nazari et al., 2022)

Hanif, Mughal, Khan, Almujally, Kim and Cha (2024) presented two CNN models which are Efficientnet-b0 and Darknet19. The components of Efficientnet-b0 include an 18-block convolutional layer, an input layer with dimensions of 224 by 224 by 3, a global mean pooling layer, a fully connected neuronal layer, and an additional layer. The Efficientnet-b0 model trained on the ImageNet dataset with 1000 classes was first implemented. This CNN model is fine-tuned by removing the last fully connected layer and adding more dense layers as it was trained using the transfer learning approach. Figure 2.7 represents the overall design of the Efficientnet-b0 model. The 19-layer Darknet19 structure is a compressed variant of Joseph Redmon's original Darknet design. With the exception of the fully connected layer, the Darknet19 architecture consists of 19 layers. 11 of these layers are convolutional. With fewer layers, it is more computationally efficient and quicker to train using the same basic structural pieces as the bigger Darknet53 architecture.

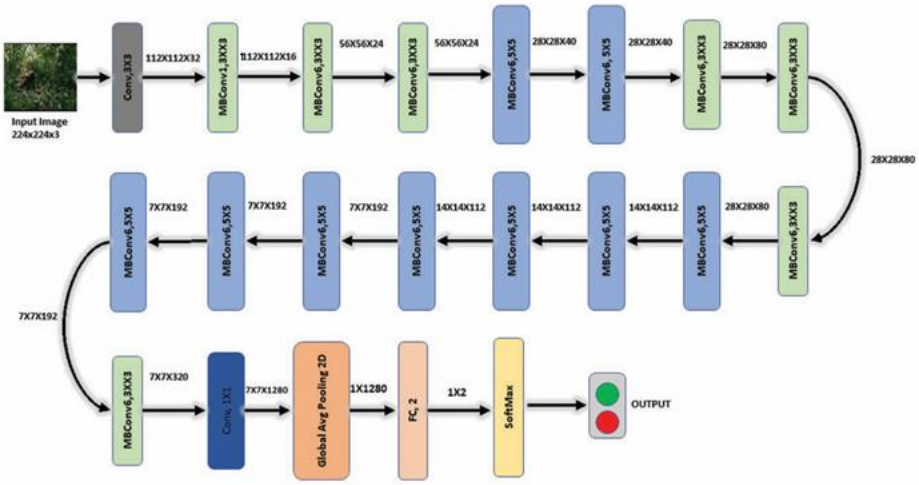


Figure 2.7: Efficientnet-b0 architecture (Hanif et al., 2024)

2.7 Datasets for Gait Recognition

Yao, Kusakunniran, Wu, Zhang, Tang and Yang (2021) suggested using the CASIA Gait Database B as the dataset for their gait recognition studies. This dataset includes 124 various individuals' gait videos taken from 11 distinct angles of view 0°, 18°, 36° till 180°. Each individual has ten patterns recorded which are two sequences with a long coat, two sequences with a backpack, and six regular sequences. As a result, this information can be used to study how various wearing styles and angles affect gait recognition. Every frame in CASIA Gait Dataset B contains only one person and the sequences are gathered in a completely supervised indoor setting.

Zhu, Guo, Yang, Huang, J., Deng, Huang, G., Du, Lu and Zhou (2021) presented the first study addressing unresolved issue which is the Gait REcognition in the Wild (GREW) benchmark. GREW contains 128,671 sequences and 26,345 subjects captured by 882 cameras in wide-open settings using real-world video captures. 233,857 sequences formed the very first set of distractors in the gait study industry. There are different types of input data such as two-dimensional and three-dimensional human poses, GEIs, optical flow, and silhouettes available in this dataset through automated process. Comparing this GREW dataset to the widely used CASIA-B and OU-MVLP dataset, GREW is more difficult to identify individuals by gait because the raw data is recorded in real settings.

Li, Makihara, Xu, Yagi, Yu and Ren (2020) assessed two datasets which are OU-MVLP and CASIA-B. With a variety of view options, OU-MVLP has become the largest database. With 14 viewing angles starts from 0° till 270°, there are 10,307 subjects. There are two sets which are #00 and #01 for every subject. Sequences #00 and #01 have been identified accordingly as galleries and probes. Another commonly used dataset is CASIA-B. There are 11 different view angles for the 124 subjects which are of range 0° to 180°. There are 10 sequences per perspective for each subject. Two sequences hold a bag, two are dressed in a coat and six are in the typical walking state.

Içik and Ekenel (2021) conducted experiments on the OU-ISIR Large Population dataset which has more than 4000 people. Two gait video sequences such as a collection of images and a probe are all included in this dataset. A concurrent recording of a gait sequences was made from four separate angles of 55°, 65°, 75°, and 85°. This dataset includes the beginning and final frame values of each of the four groups of each video sequence which are called View-55, View-65, View-75, and View-85. Another dataset used is CASIA-B gait dataset where each of the 124 subjects is captured from eleven varied points of view that includes 18° constant intervals. There are 10 video sequences for every perspective for every subject. There are about 14,000 video sequences in the CASIA-B dataset due to 124 individuals × 11 viewpoints × 10 videos.

Kumar et al. (2021) evaluated CMU MoBo dataset which has six distinct perspectives that are used to see silhouettes. In every situation when choosing between width or height signals is necessary, the signal with the appropriately greater amplitude must be used. In order to provide suitable volume recognition, this dataset was used to simulate the frontal depth of GEV. 25 individuals are included in the dataset under four different testing groups. Six cameras concurrently record each time an individual uses an indoor treadmill for walking. There are different groups such as walking on a sloped surface, walking while holding a ball, walking quickly, and slow-paced walking.

2.8 Performance Evaluation Metrics

Sayeed et al. (2022) demonstrated that the suggested method for identification performance is assessed using the CCR which is particularly the Rank 1 identification

rate. The formula of CCR is the sum of TP and TN divided by the sum of all the predictions which are TP, TN, FP, and FN. Kececi et al. (2020) explained that TP is when the positive class is predicted by the model with high accuracy. As for TN, the negative class is predicted by the model.

Sayeed et al. (2022) stated another calculation which is EER are also used to evaluate how well the verification task which is the one-to-one matching performed. EER is the rate at which FAR and FFR are equal in order to get the optimal threshold value. This rate should be zero for optimum efficiency in the gait verification system where the performance is better when the rate is lower. FAR which sometimes referred to as Type-I error is an error rate that occurs when a fake person is incorrectly assumed to be the real one. To lower the FAR error, the threshold value must be increased. The Type-II error rate which is FRR is the rate that occurs when an appropriate person is mistakenly turned away. The FRR error will rise as a result of the increased threshold value.

Şahan, Sahin and Gülağız (2024) researched a performance evaluation criterion which is the CMC. CMC is frequently used in machine vision and biometrics especially for tasks involving recognition systems like gait recognition. The efficiency of assessing the accuracy of identifying systems may be better. The CMC curve effectively shows the possibility of an input identity appearing in the top K rankings of a sorted list of possibilities generated by the system as this CMC is a rank-based statistic. Although the CMC curve is an effective method for evaluating identification systems, it does have some drawbacks is completely ignores the overall accuracy and trustworthiness of outcomes as it focuses on ranking efficiency. However, the initial point on the CMC curve is regarded as the rank-1 recognition accuracy.

Raihen & Akter (2024) evaluated effectiveness of correct classification rate or also known as accuracy which includes four variables such as TP, TN, FP, and FN. These four variables also form tables that are used to explain how a classification model performed on a collection of previously collected test data which is known as confusion matrix. Table 2.3 below shows confusion matrix in its format. There are also various measures employed for evaluation including TP, FP, Precision, Recall, F1

score and ROC area. Slemenšek et al. (2023) stated that the number of instances successfully categorised by the algorithm is denoted by the TP and TN variables whereas the number of instances wrongly classified is denoted by the FP and FN variables. The goal is to maximise the rates of TP and TN while reducing the rates of FP and FN.

Table 2.3: Confusion Matrix Format (Raihen & Akter, 2024)

	Predictive Positive	**Predictive Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

2.9 Summary

Table 2.4: Summary of Gait Recognition Based Studies

Author (Year)	Algorithm	Dataset	Accuracy
Zhu et al. (n.d.)	GaitSet	CASIA-B	84.48%
Kececi et al. (2020)	IB1 Random Forest Bayesian Net	HugaDB	99.00% 99.00% 99.00%
Zou et al. (2020)	CNN and LSTM	whuGAIT	97.33%
Içik & Ekenel (2021)	VGG16	OU-ISIR CASIA-B	95.00% 100.00%
M. H. Khan et al. (2021)	Cubic SVM	CASIA-B	98.20%
Saleem et al. (2021)	Cubic SVM	CASIA-B	97.30%
X. Li et al. (n.d.)	Ensemble	OU-MVLP CASIA-B	96.80% 98.60%
Yao et al. (2021)	GEI and SGEI	CASIA-B	100.00%
Asif et al. (2022)	SVM	CASIA-B	83.33%
Chao et al. (2022)	GaitSet	CASIA-B OU-MVLP	100.00% 90.70%
H. Li et al. (2022)	GaitSlice	CASIA-B OU-MVLP	99.70% 91.60%
Nazari et al. (2022)	Random Forest Gaussian Naïve Bayes LDA Decision Tree Deep CNN	EMG	75.00% 73.00% 73.00% 72.00% 79.00%
Sayeed et al. (2022)	CNN with LeakyReLU	OU-ISIR OU-MVLP	93.84% 56.80%
Sharif et al. (2022)	OaA-SVM	CASIA-B	95.75%
Narayan et al. (2023)	CNN Capsule Network RNN GAN DBN CNN + LSTM	CASIA-B	98.90% 84.50% 96.15% 93.20% 93.00% 99.60%
Sepas-Moghaddam & Etemad (2023)	3D CNN	CASIA-B OU-ISIR OU-MVLP	90.40% 99.90% 89.18%
Hanif et al. (2024)	Efficientnet-b0 and Darknet19	CASIA-B	100.00%

The ability to recognise a person's steps is an important biometric aspect of person authentication. Unlike other biometric technologies such as iris scanning, fingerprinting or facial recognition, gait analysis does not need individuals to actively

engage and it is safe. Individuals with this characteristic tend to be more cautious or anxious. It differs from previous biometric approaches in that it may be used for gait recognition remotely. Although gait analysis may be inaccurate due to issues such as camera angle, lighting, and clothing choices, it is still fairly challenging to recreate another person's steps.

Machine learning and deep learning have revolutionised gait recognition by enhancing the extraction of features, classification, and analysis. Models like SVM and Random Forest improve results by hyperparameter adjustments while traditional techniques utilising algorithms such as HOG, PCA, and DTW concentrate on analysis of movement and reduction of dimensionality. By employing measures like the geometrical distance, template-based methods like GEI and CGI are able to construct stable, view-invariant gait models. Improvements in gait phase prediction, reduced overfitting, and effective feature extraction in space and time have all resulted from deep learning's use of CNN. Improvements in spatial-temporal features and frame selection are achieved by the integration of complex elements in innovations such as the GaitSlice model. A combination of these developments shows that current methods are successful in obtaining trustworthy gait recognition.

The gap in this study is the fusion approach should be improved to enhance the performance of gait recognition system. Additionally, gait recognition should be applied and assess through the use of simulation platforms and robotics in the real-world application. There are various methods that may produce better results and discover new opportunities in various fields of study.

CHAPTER 3

METHODOLOGY

3.1 Overview

This work demonstrates that pre-trained ViT which is a gait recognition using a pre-trained Vision Transformer that recognise patterns of gait by extracting both temporal and spatial data. The research framework involves several phases with the initial phase is obtaining raw silhouette sequences from available datasets. The raw silhouettes are then pre-processed to GEI which involves alignment and normalisation to ensure the data is compressed into a single frame. The main component is fine-tuning the pre-trained ViT (vit_base_patch16_224). Classification is done to classify individuals based on their gait through the customisation of classifications head such as three-layer MLP. This model is then evaluated using the classification metrics like accuracy, precision and many more to measure the performance of model in different situations and conditions. PyTorch is implemented on a hardware setup with NVIDIA GPU to ensure that the training process is smooth and efficient. This project is done within a structured timeline with several phases.

3.2 Research Framework

The are a few important steps in this project. The first phase is gathering data, which record the spatio-temporal aspects movement of a person. These greyscale images is a solid representation of gait patterns as it has several conditions like normal walking, wearing clothes and carrying bags. These conditions are captured in several different viewing angles. It also involves data normalisation and augmentation through preprocessing, feature extraction and classification. The framework end with an evaluation step to analyse the model functionality and dependability to ensure adaptation to different situations. Figure 3.1 shows the Gait Recognition Framework.

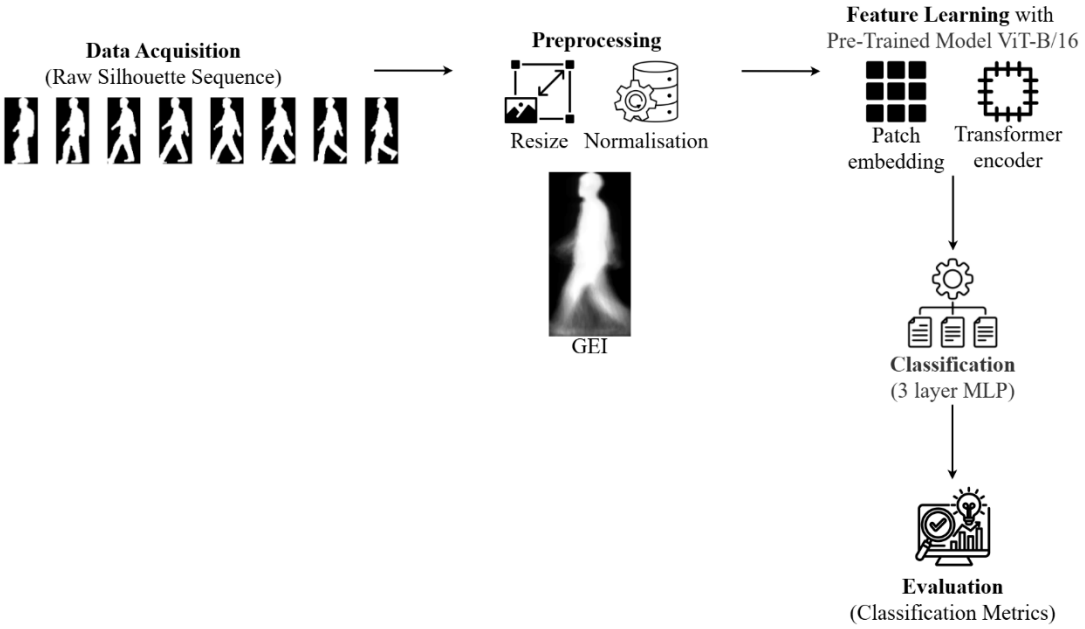


Figure 3.1: Gait Recognition Framework

3.2.1 Data Acquisition

The first step in a comprehensive research framework is gathering data such as raw silhouettes sequences in CASIA-B datasets which is a large multi-view gait dataset (Song et al., 2024). These silhouettes represent individuals in binary two-dimensional at every phase of a gait cycle in which there are a total of 124 individuals. The dataset consists of individuals with several walking conditions such as normal walking, wearing different types of clothes, and carrying bag in eleven different viewing angles. The range of angles are 0° to 180° with an interval of 18°. There are a total of ten sequences such as six normal walking sequences, two sequences of walking with a coat and two sequences of walking with a bag for every individual.

The human silhouettes are extracted from video files. The filenames for the silhouette in CASIA-B dataset has format of 'xxx-mm-nn-ttt.png', where each component represent a thing. The video files are processed to extract the silhouettes of humans. The structure for the silhouette filenames in CASIA-B dataset is 'xxx-mm-nn-ttt.png', where each component represents an element. The representation of the components is shown in Table 3.1 below:

Table 3.1: Representation of Component

Component	Element
xxx	subject id, from 001 to 124
mm	walking status such as: • 'nm' which is normal • 'cl' where the individuals are wearing a coat • 'bg' where the individuals are carrying a bag
nn	sequence number
ttt	viewing angle, such as: • '000', '018', ..., '180'

3.2.2 Preprocessing

The following step is preprocessing which makes the data ready for the analysis process and convert the raw silhouettes into Gait Energy Image (GEI). The main function is calculating the centre of mass or better known as centroid of the silhouette image. This function is to improve the alignment for analysis of gait. It is done by computing the vertical and horizontal projection along each row and column respectively through the process of averaging the values of pixel. It also calculates the weighted average positions to get the accurate centroid value and return the integer coordinates or floating-point values.

The arrangement and normalisation of all gait patterns to a uniform size, position, and dimension is an important part of this process. At this phase, any problems related to the extraction process of silhouette, such as different sizes or missing images are resolved. The output images which is the GEI has same dimensions which makes the size standardise. As for the position, it is normalised to ensure that it is centring the silhouettes. It also cleans the raw data where it checks and handles if there are missing or empty images.

The GEI is calculated by aligning and averaging the silhouettes from a whole gait cycle. In a two-dimensional greyscale image, where pixel intensity represents the raw silhouette, this step records the temporal and spatial aspects of gait. Preprocessing ensures the transformation of raw silhouettes to GEI are uniform to reduce the

complexity of computation for duplicated data. It also minimises the visibility of any distortions in the silhouettes. Figure 3.2 shows the GEI for normal walking, wearing a coat and carrying a bag condition.

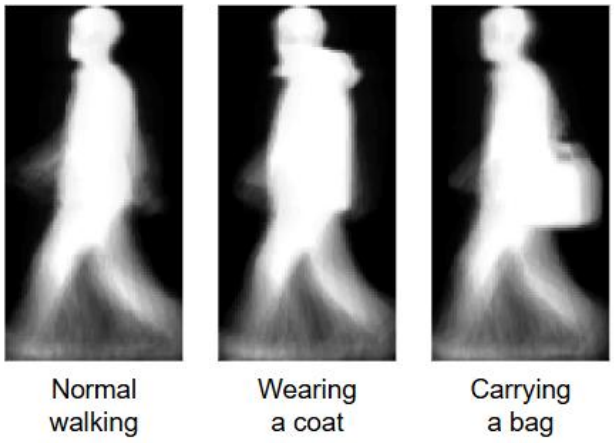


Figure 3.2: GEI for all three conditions

According to (Gul et al., 2021) the formula for GEI is shown below where *N* is the total frames in a gait sequence. $F_t(x,y)$ is the value of silhouette at *t* frame where *x* and *y* are the coordinates of 2-dimensional image. $G(x,y)$ is the final output of GEI.

$$G(x,y) = \frac{1}{N}\sum_{t=1}^{N} F_t(x,y) \tag{3.1}$$

3.2.3 Feature Learning with pretrained model ViT-B/16

The feature learning process involves the use of method like Transfer Learning with Vision Transformer (ViT). vit_base_patch16_224 is used as it is pretrained on ImageNet so it is already familiar with the general characteristics of images such as edges, textures and forms (Dosovitskiy et al., 2020). It extracts hierarchical visual characteristics automatically through patch embedding like 16×16 patches. It also uses multi-head self-attention layers to recognise long-range dependencies which is important in identifying movement, position, and form. It refines features across layers along with layer normalisation and residual connections.

3.2.3.1 Architecture of pre-trained ViT

The goal of the pre-trained ViT architecture is to efficiently analyse and identify human gait patterns. It modifies several parts of the generic Vision

Transformer model and combines them to meet the unique requirements of gait recognition. In this pre-trained ViT, the process of normalisation and alignment are preprocessing steps which require to make input GEI images more consistent.

The next task is to divide the images in the form of fixed-size patches and embed them into an area with higher dimensions. The algorithm is able to capture overall connections across patches because positional embeddings are implemented to maintain spatial data. Then, the enhanced patches undergo processing using multiple layers of multi-head self-attention, feed-forward networks, layer normalisation and residual connections inside the architecture. This allows the model to accurately acquire and represent gait features for identification purposes. Figure 3.3 shows the architecture of pre-trained ViT.

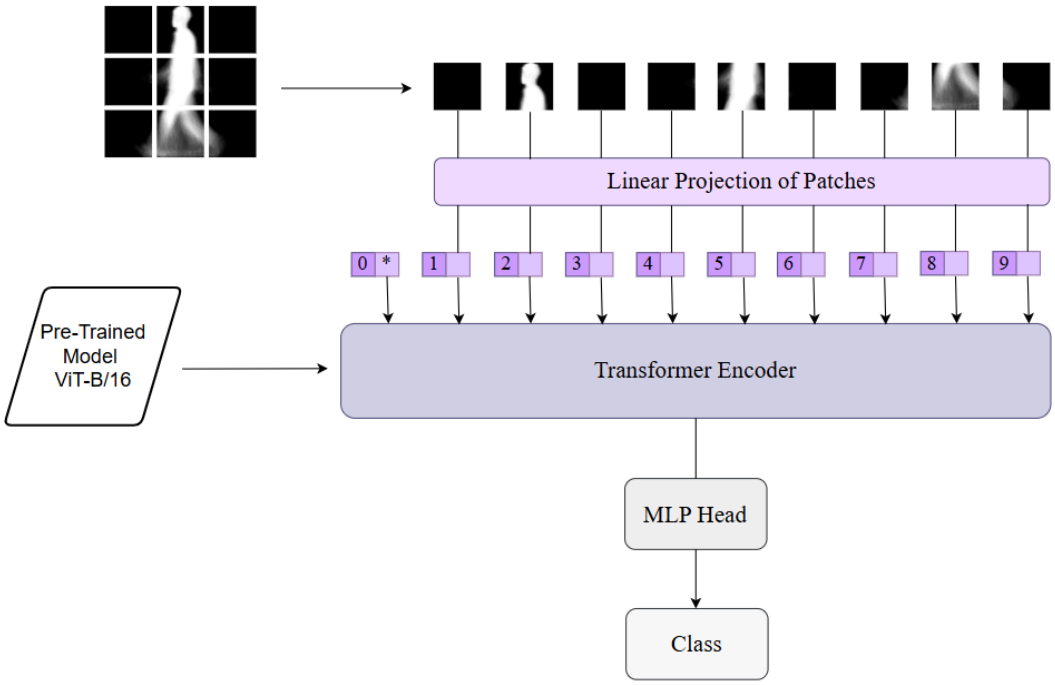


Figure 3.3: Pre-trained ViT-GR Architecture

Firstly, it involves input processing where an input GEI image which has the dimensions of height, width and channels, $H \times W \times C$ is divided into smaller and not interconnected patches with dimensions $P \times P \times C$. A one-dimensional vector is created by flattening each patch, which produces patches in sequence with the formula (Dosovitskiy et al., 2020).

$$N = \frac{H \times W}{p^2} \tag{3.2}$$

The patches are projected linearly into an embedding space with higher dimensions through the use of a learnable weight matrix *W*. A series of patch embeddings is formed when this process transforms every single patch into a dimension vector *D*.

Each patch embedding has a positional embedding included in it so that the information regarding the placement of patches is included in the image. Since the transformer does not have the natural spatial structural knowledge seen in convolutional layers, these embeddings assist the model in understanding the exact placements of patches in the initial image.

Next, is the integration of a unique token known as classification token is added to the input sequence along with the patch embeddings. This token is essential since it collects data from every patch in the layers of transformer. When doing tasks related to classification, this token acts as a universal representation of the full image by using the self-attention process to draw attention to the various patches.

The transformer encoder receives the resultant sequence, which contains the class token as well as the patch embeddings that include additional positional encodings. It helps to efficiently acquire gait characteristics from GEI or other gait representations via learning. This serves as the input for the final estimation at the subsequent processing steps.

In addition, the transformer encoder which has Multiple layers of Multi-Head Self-Attention (MHSA), Feed-Forward Networks (FFNs), layer normalisation and residual connections which forms the fundamental structure of the ViT are assembled in the transformer encoder.

These components are used to analyse the sequence of input and give a representation that the model can interpret. An embedded or ongoing representation of

the given input sequence is produced by the encoder after processing the input sequence. The diagram of the transformer encoder is shown in Figure 3.4.

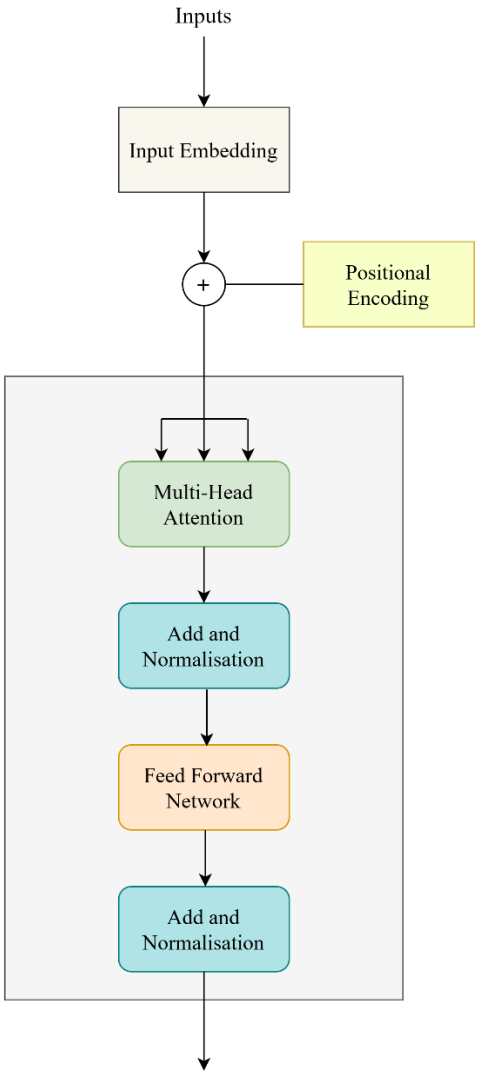


Figure 3.4: Transformer Encoder

The MHSA involves projecting every single patch embedding in the ordered set into one of three matrices such as Query Q, Key K, and Value V. The learnable weight matrices used for this purpose are W_Q, W_K and W_Y. By utilising scaled dot-product attention, the attention mechanism calculates relationships between patches with the formula (Zhou et al., 2024).

$$Attention\ (Q, K, V) = Softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{3.3}$$

This process lets the model concentrate on key areas of the image by assigning importance to each patch. The focus in every head is shown in the formula below. Each of the three heads uses its own version of Q_i, K_i, and V_i to represent Q, K, and V to implement the attention computation for various parts of the process of learning (Zhou et al., 2024).

$$Head_i = Attention\left(Q_{i,}K_{i,}V_i\right) \tag{3.4}$$

To generate the last output for that layer, the outputs from several attention heads are combined and passed through linear transformation. The computations performed by the attention module are executed in parallel. As a projection matrix, W contains the combined outcomes of all heads as shown in the formula (Zhou et al., 2024).

$$MHSA\ (Q, K\ , V) = Concate(Head_{1,\dots,}Head_n)W \tag{3.5}$$

The result is then sent through a network of feed forwards after the MHSA phase. A multi-layer perceptron, also known as MLP, is a type of network consisting of two completely interconnected layers that can differentiate input that is not linearly separated. It is composed of fully connected neurones with non-linear activation functions that are organised in layers such as the Gaussian Error Linear Unit (GeLU). Positive or negative values for the GeLU function result in a more curved function that is better able to estimate complex functions (Peltekis et al., 2024).

$$GELU\ (x) = x\ \cdot\frac{1}{2}\ [1 + \text{erf}(\frac{x}{\sqrt{2}})] \tag{3.6}$$

Later, layer normalisation and residual connections are used after MHSA and FFN to stabilise training and enhance gradient performance. The issue of diminishing gradients can be solved through Residual Connections, which are also called skip connections. The gradient can go directly across the network without going through any intermediate layers. The learning performance of the networks will be improved as the gradient may move across the layers with less information loss.

Layer Normalisation is a method that ensures inputs remain between a specific range throughout training. The average and standard deviation of the inputs for every given sample are computed to complete this function. The inputs are subsequently

transformed to a standard distribution by a normalisation process. This normalisation phase will ensure a consistent and efficient training process regardless of the complexity and scalability.

Finally, the classification head is where the transformer extracts multidimensional characteristic and requires extra modifications in order to have the data ready. The classification head, which is MLP, a fully connected neural network layers that maps feature of high dimensional to the total number of classification classes. In order to construct the distribution of probabilities across the classes, this head uses the output that corresponds to the token. This unique token is used to gather data from the whole sequence which is from the transformer in tasks related to classification of image.

3.2.3.2 Transfer learning

In this phase, it reuses the pre-trained model that were trained on ImageNet dataset rather than training a vision transformer model from scratch. This process helps the learning process which is used to target the predictive function. The model learns patch embeddings, self-attention across patches and classification head for pre-training tasks. This pre-trained ViT model is trained along with the transfer learning where several parameters were used and modified to suit the CASIA-B dataset.

The classification head is modified where it minimises overfitting and enhance class separability by changing the original head for a deeper and regularised one. It is replaced with a 3-layer MLP such as the first layer has a dimensionality of 1024, batch normalisation, GELU activation and dropout of 0.3. The second layer has dimensionality of 512, batch normalisation, GELU activation and dropout of 0.2. The final layer is the output for number of classes.

This transforms the dimensionality by taking the input features and projects into the dimensional space. As for the batch normalization, it speeds up and stabilises training because it prevents expensive pre-training from scratch. It also minimises internal variables shift by activation normalisation. The GELU has better performance

in attention-based models as it matches the implementation of a Vision Transformer. While maintaining effective accuracy, this phase has the flexibility to fine-tune tasks based on classification.

3.2.4 Classification

The following phase is using the extracted features to classify the individuals or distinguish between gait patterns. This pre-trained ViT is used to identify and retrieve important patterns from the GEI though the learning process. These traits serve as inputs for additional classification and are important for differentiating individuals from one another according to their gait. This method not only makes it easier to distinguish between people based on their gait, but it also ensures stable and effective training.

The GEIs characteristics such as the silhouette shape and patterns through the analysis of patterns across images are transformed. However, it still maintains the important gait characteristics such as structure and positional details. This pre-trained ViT-GR is suitable for classification task because it can handle gait patterns in an efficient manner where the customisation for classification head improves the features accordingly. Therefore, it is relatively easy to identify individual based on their gait.

3.2.5 Evaluation

The evaluation process examines the efficiency and dependability of the model. Testing the system in various situations and conditions is a common way to assess the effectiveness of model and its robustness. The strengths and limitations are the information gained during the assessment process. This is important for future enhancements to classification algorithms. In this project, there are several evaluation components used to evaluate the performance of this pretrained vision transformer model.

Test loss is calculated to verify the method of changing a model's settings to make sure its predictions match what is observed. Lower test loss means the method

is relevant and suitable as it shows the degree of dependability and precision in the predictions made by the model.

The accuracy is also calculated to determine the number of correct predictions out of total predictions. The formula below shows the calculation of accuracy (Slemenšek et al., 2023).

$$Accuracy = \frac{TP+TN}{TP+FN+FP+TN} \tag{3.7}$$

F1 score is also calculated to find out if the class is imbalance. The formula below shows the calculation of F1 score (Slemenšek et al., 2023).

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision+Recall} \tag{3.8}$$

Precision is used to calculate how many correct predictions were made out of all the predicted ones. The formula below shows the calculation of precision (Slemenšek et al., 2023).

$$Precision = \frac{TP}{TP+FP} \tag{3.9}$$

Recall is used to calculate how many predicted ones out of all true identities. The formula for recall is shown below (Slemenšek et al., 2023).

$$Recall = \frac{TP}{TP+FN} \tag{3.10}$$

As for confusion matrix, it is used to show actual and predicted classes to identify which classes are often misclassified. The row and column represent true class and predicted class respectively. Classification report is used to check the effectiveness in recognising each individual's gait by combining precision, recall and F1-score for each class.

There are few visualisation graphs displayed to show the performance of model such as training and validation loss over epochs which helps to check if the model is

overfitting or underfitting. This graph plotted to guide the process of optimisation where the loss represents the differences between predicted output of model and actual target value. The model try to minimise loss by adjusting the weights during training process. The training loss refers to error on the training data where validation loss is the error on unseen data that is used to evaluate the performance of model.

As for the graph of training and validation accuracy over epochs, it is used to view the performance of model during progress of training. This graph is used to understand how well the model learns from training data and generalise to unseen data. The training accuracy refers to the accuracy of training data over the iterations of epochs whereas the validation accuracy is accuracy on another validation dataset that is unseen during training process.

3.3 Software and Hardware Setup

This project runs on Windows Operating System and it has several software packages used. This project is developed and carried out in Jupyter Notebook using python version 3.11.11. Anaconda Navigator is used to access the environment that has PyTorch installed as it is a graphical user interface. Users can find the environments and packages easily as compared to using command prompt. The PyTorch version is 2.6.0 and it is built with CUDA 11.8. PyTorch is used in this project as it is flexible and easy to use where it simplifies the computation that is necessary for training of Vision Transformer in this project. As for CUDA, it is supported to ensure that the computations on GPUs are efficient and compatible.

The core library used in this project is the PyTorch which contains neural network layers, loss functions and optimisers. The torchvision.transforms preprocess the transformations of image data. It also ensures that the data loading and batching are efficient and suitable for training process. The datasets are also split into training, validation and test sets using the random split function. The torch.utils.data.WeightedRandomSampler helps to handle class imbalance by oversampling underrepresented classes. It also simplifies the loading process of image datasets as each class has its own folder.

There are several libraries used in the software such as the os library which is used to interact with the operating system such as the operations of file directory. The numpy library is used for operating numerical elements on arrays as it is useful in handling image data. The matplotlib.pyplot module is used to plot graphs and visualise data such as the training and validation loss along with accuracy. As for the cv2 which is an OpenCV module which is used for image processing tasks like reading, resizing and preprocessing the images.

The sklearn.metrics is an evaluation module used for classification which consists of accuracy_score, precision_score, recall_score, f1_score, classification report and confusion matrix. The timm which is PyTorch Image Models is a library with pre-trained models like the Vision Transformers (ViT). As for tqdm, it is a library for displaying bars for loops and iterations in a simple way. The collections.Counter is used to count the frequency of elements in a list such as the distribution of class in datasets. Lastly, the time module is used to measure the execution period and identify if there are any delays. As for the datetime.timedelta, it is used to format durations like the duration of epoch training process.

As for the hardware, this project is created and the pre-trained vision transformer model is trained on a laptop with Core i5 9th Gen CPU, NVIDIA GeForce GTX 1650 GPU, 16GB RAM and 4GB GPU memory. This allows the performance of training using lower power consumption as it increases the speed of model training as compared to training on CPU. The 16GB RAM helps to handle datasets that have moderate size. It also supports preprocessing tasks that are slightly complex to ensure there is no issue happening outside of the GPU memory. To conclude, these specifications are suitable for training Vision Transformers on smaller scale without the need for expensive GPUs.

3.4 Project Timeline

3.4.1 Phase 1

Figure 3.5 shows Gantt chart that present each task and duration they take for a 13-week project in Phase 1. Weeks 1 to 3 are assigned to the literature review. Weeks

3 to 5 for the introduction. Weeks 5 to 7 with regard to the methodology. Weeks 8 to 10 spend on experimental design. Weeks 10 to 12 have been allocated to the conclusion and Week 13 for the presentation. The time allotted to each work is shown to see how the project is progressing.

Tasks	Week												
	1	2	3	4	5	6	7	8	9	10	11	12	13
Chapter 2: Literature Review													
Chapter 1: Introduction													
Chapter 3: Methodology													
Chapter 4: Experimental Design													
Chapter 5: Conclusion													
Presentation													

Figure 3.5: Gantt Chart

3.4.2 Phase 2

Figure 3.6 shows Gantt chart that present duration of each task in a 13-week project in Phase 2. Weeks 1 to 6 are assigned to the training and testing of models which involves the implementation of coding process. The methodology is done during Weeks 7 to 8 and the experimental design is done in Weeks 8 to 10. Weeks 10 to 12 have been allocated to the conclusion and Week 13 for the presentation. The period for each task is shown to see how the project is progressing and completed within the period of time.

Tasks	Week												
	1	2	3	4	5	6	7	8	9	10	11	12	13
Training and Testing of Model													
Chapter 3: Methodology													
Chapter 4: Experimental Design													
Chapter 5: Conclusion													
Presentation													

Figure 3.6: Gantt Chart

3.5 Summary

In summary, this pre-trained Vision Transformers is suitable for gait identification as it is used to extract spatial and temporal data from raw silhouette images. The primary steps are including data acquisition of raw silhouette sequences. Preprocessing is used to generate GEIs through alignment and normalisation to get the

data in single frame. Feature learning and classification are done in enhancing the model to be suitable for recognising the gait patterns of every individual. Evaluation is used to determine the performance through metrics and confusion matrix. The framework shows it ability to adapt to different walking conditions at several viewing angles. This project is developed in PyTorch with CUDA support, NVIDIA GTX 1650 GPU, 16GB RAM to shorten the training process. This project follows a relevant timeline which is split into two phases, Phase 1 and Phase 2.

CHAPTER 4

EXPERIMENTAL DESIGN AND ANALYSIS

4.1 Overview

This experimental design is implementing pre-trained ViT based on CASIA-B dataset that includes various walking sequences of individuals in different settings. The situations included normal walking, walking while wearing clothes and walking while carrying a bag in various viewing angles. The dataset contains wide range of situations where 124 human silhouettes are extracted from the video data which is the primary component of this analysis. Preprocessing steps are done to extract GEI which is a representations of gait cycle in one image. This is done to prepare data for model to train where centroid calculations, area extraction and normalisation are implemented to ensure the dimensions are uniform. Several data augmentation techniques such as random cropping, flipping, rotation and many more are also applied to avoid overfitting. The experimental design consists of three experiments conducted to evaluate the performance of model under different conditions. The experiments are normal walking sequences for both training and testing, mixed-condition training and individual testing and lastly, full combination of condition for both training and testing. Comparative analysis are also conducted for each experiment and it is proven that this pre-trained ViT outperform most of the existing models based on its condition.

4.2 Dataset

An extensive multiview gait database which is CASIA-B was established in January 2005 (Song et al., 2024). The gait data was collected from eleven different angles and involves 124 individuals. The available viewing angles range from 0° to 180°, with an interval of 18°. There is a total of ten gait sequences per individual over eleven views, with six normal walking sequences, two coat walking sequences, and two bag walking sequences. Normal walking, with coat on, and carrying a bag condition are the three differences that are evaluated individually.

4.3 Data Preprocessing

There are several preprocessing steps done to extract GEI from the raw silhouette sequences obtained from the CASIA-B dataset. The first function is mass_center which is used to calculate the centroid of a silhouette image. It is done by taking an input silhouette image and calculating the vertical and horizontal projections through the process of averaging the value along the X and Y axis which is the horizontal and vertical axis respectively. The weighted average position is then calculated using these projections to get the centroid. After calculating, it returns the coordinates either in rounded integer or floating-point value. The rounded integer is set by default.

In addition, the second function is image_extract which is used in input image to extract and normalise the silhouette area. The non-zero area such as the columns and rows are searched. This is the region that has the silhouettes and bounding box are used to identify them. The bounding region is then centred horizontally around the centroid of silhouette to crop the image and resized to the dimensions wanted using bicubic interpolation. Empty images are handled by returning a zero array.

The last function is generate_geis_for_all_subjects which is used to generate GEIs for all the 124 individuals under the three different conditions. It creates a necessary directory for output produced and arrange it according to subject ID. Later, it processed images for each subject and condition by loading all the raw silhouette images and apply preprocessing steps to every image. The GEI are calculated by averaging all pixel values across all frames over time and saved as PNG file in the output folder.

The raw sequence of the silhouette images for the condition of normal walking, wearing a coat and carrying a bag are displayed along with the generated GEI. However, it is set to only display eight raw sequences for every individual. Figure 4.1 below shows the eight raw sequences and generated GEI for first individual with normal walking condition. Figure 4.2 below shows the eight raw sequences and generated GEI for 62nd individual with carrying a bag condition. Figure 4.3 below

shows the eight raw sequences and generated GEI for 124th individual with wearing a coat condition.

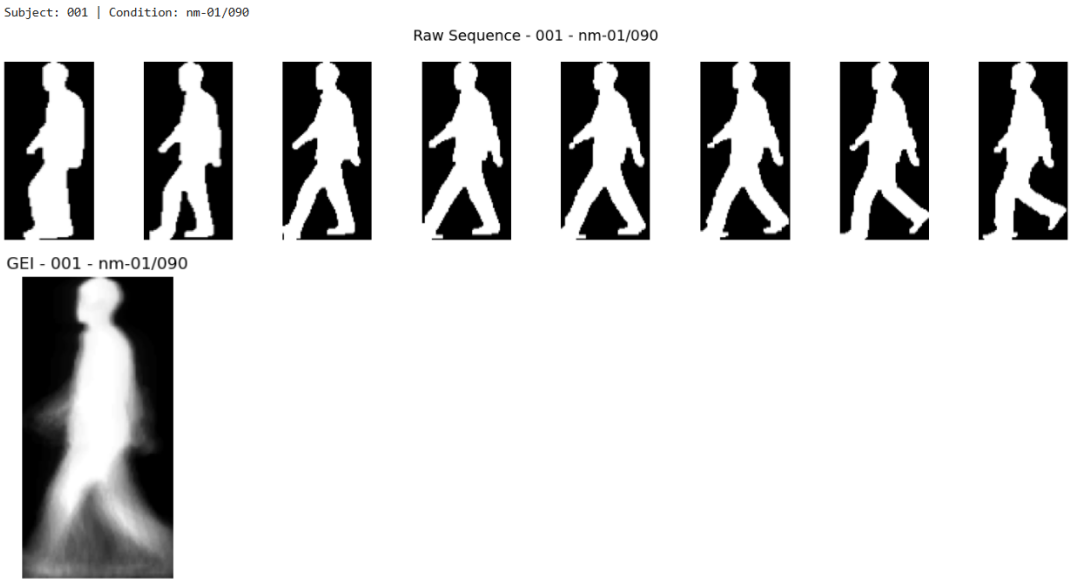


Figure 4.1: First individual with normal walking condition

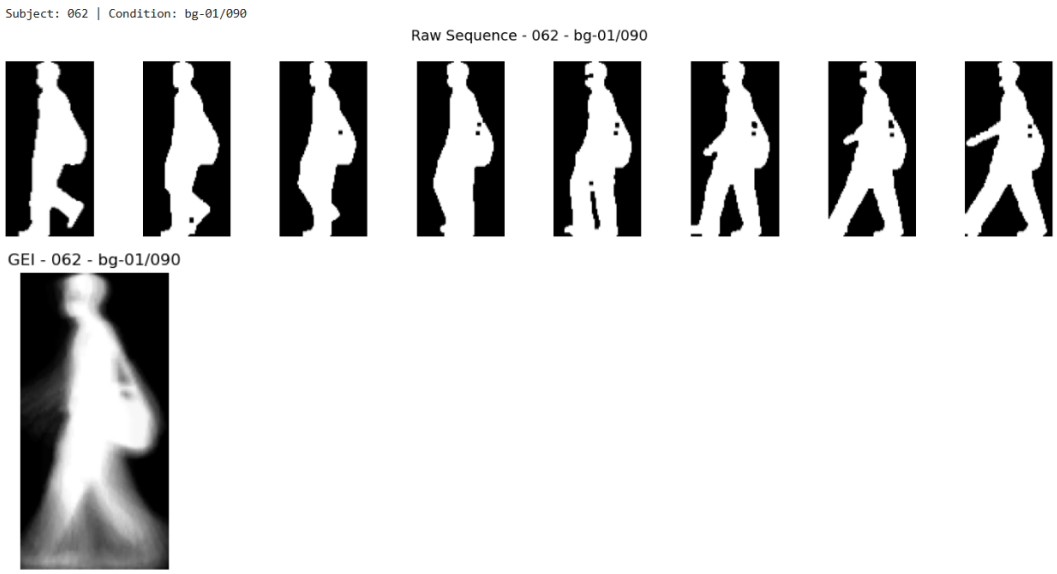


Figure 4.2: 62nd individual with carrying a bag condition

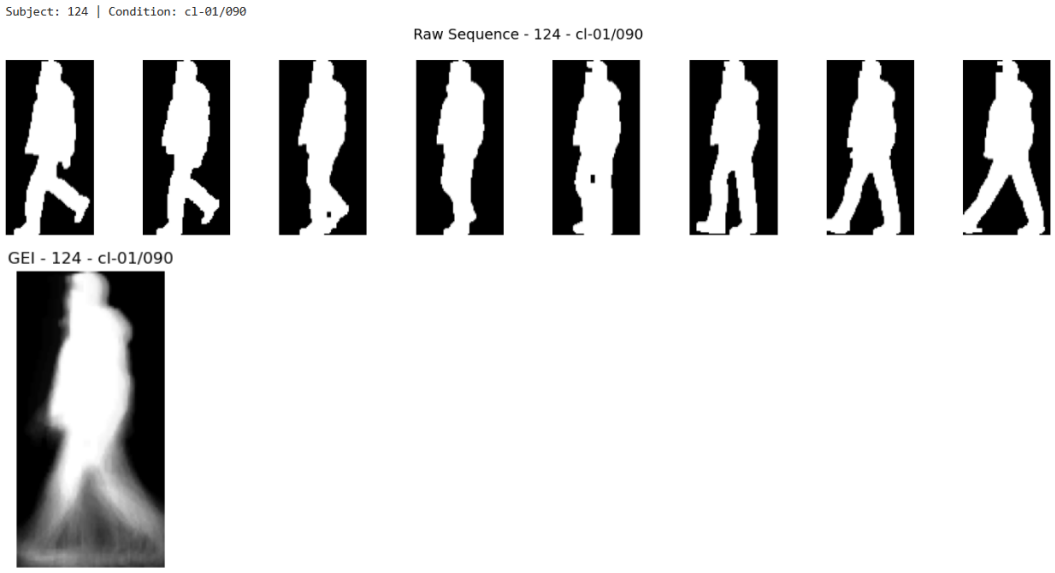


Figure 4.3: 124th individual with wearing a coat condition

As for data augmentation, transformation is done to artificially increase the diversity of training dataset by creating another copy of dataset with modifications. This is done to ensure the model can generalise better and improve the performance if it deals with limited data or potential overfitting issues. There are several augmentations for training transformation such as grayscale with three output channels which is used to convert single-channel GEI to three channels as the ViT is pre-trained with inputs of 3 channels which makes it a requirement.

Next, the data is resized and random crop to make it suitable for data in different locations and maintain the effectiveness. Random horizontal and vertical flip is also implemented so it can create walking direction in mirror-image form. Random rotation is also used to make different variations in walking angles whereas random affine is used to add translation, rotation and scale noise based on the gait angle variation. Although the GEI are grayscale, colour jitter is used to add brightness and contrast noise to enhance the appearance of the data.

Furthermore, random perspective improves the generalisation and robustness of model by diversifying the orientations. Random erasing is used to randomly select a rectangular region and erases its pixels with random values to decrease risk of overfitting and simulates occlusion or blockage. Finally, normalise is used to standardise the value of pixels for the pre-trained weights. As for the testing

transformation, it only involves resizing and normalising without any randomness to ensure consistency.

4.4 Model Training

This training process is the important part of the model as it needs to learn to classify correctly based on the input by adjusting its parameters. The structure of the training loop is the number of iterative processes which is known as epochs are set to 100. It runs up to 100 epochs unless the early stopping function is triggered. This is because the model sees whole training dataset multiple times depending on the number of epochs. The model learns from training data and evaluates its operations on unseen data. This is to ensure that is does not overfit and achieve satisfactory performance.

Moreover, the batch processing involves the training data processed in batch of size 64 to ensure that it is trained step by step rather than individually. This helps to provide more updates on frequently used parameters as well as balances the usage of memory and stability of gradient. It also enables training on large datasets that do not fit in memory as some of it are too complex to be trained. Next, automatic mixed precision training is done to speed up training process and decrease usage of memory on CUDA-enabled GPU. This enables some operations to execute in float16 which is half precision rather than float32 which is a full precision. It ensures that numerical stability is well-maintained as it train faster without greatly impacting the accuracy.

As for the forward pass of the model, it runs in mixed precision and passes all the input images to produce predictions through the network layers. It also eases the loss computation by measuring the degree of wrongness for the predictions as compared to true labels. Besides that, the backward pass adjusts the weights and bias to minimise the cost function iteratively. The optimiser is used to reset gradients of the parameters that have been optimised before calculating new gradients for current batch. This is to ensure that the gradients do not accumulate in every training iteration and it only match the current batch.

Scale is also implemented to scales the loss computation through dynamic factor. This is done as the gradients calculated in float16 may be small value or underflow to zero which causes instability in training process due to its numerical issues. The reason behind using GradScaler is it can multiply the loss computed with large scale factor so the gradients can be in a stable range of numerical to maintain the percison of gradient. The norm 1.0 of gradient is then clipped to a maximum value to avoid overflow gradients which can lead to training destabilization. The unscaling of gradients and detection of overflow are integrated into the optimisation process to ensure safe and stable parameter updates. The scale factor is then updated to make it available for next iteration through the dynamic adjustment of scaling which optimises training speed and stability over time.

4.5 Hyperparameter Tuning

There are several hyperparameters in these experiments used for hyperparameter tuning to obtain the most suitable combination of hyperparameters to ensure the optimisation of performance of model. Firstly, the batch size is 64 which is the amount of training samples that are processed first before updating the model during training. This batch size is selected as it is a balance between the memory usage and training stability on 4GB GPU which offer correct gradients and better use of parallel computing on GPU.

Besides, the loss function implemented is CrossEntropyLoss which is used to measure the predictions if it matches the actual results as performance evaluation. This also helps to reduce the error in model and achieve optimal solution in which smaller error represents better model performance. Next, the optimiser applied is AdamW as it can separate weight decay from learning rate update or in simpler terms, the gradient step. This is to ensure better regularisation for improvement of model generalisation as it is suitable for pre-trained ViT that are trained on large datasets like ImageNet.

Furthermore, the learning rate is 0.00004 which helps to control the number of weights updated each time when the model is trained. This learning rate is suitable for this pre-trained ViT as it helps to prevent overshooting minima and achieve smoother convergence. A regularisation technique is used which is weight decay of value 0.0001

that helps to add L2 norm of weights to prevent overfitting which reduces the memorisation of training data. This value can slightly help to penalise large weights in which the model will be general and not underfit.

Moreover, the scheduler is CosineAnnealingWarmRestarts is used to ensure better convergence and prevent local minima from happening. The number of iterations for first restart is every 10 epochs so the model has new learning chances. The multiplication factor after restart is 1 and the minimum learning rate is 0.000001 to make sure that it does not reach zero. Additionally, early stopping is also implemented such as patience=10 which is used to prevent overfitting by terminating the process of training once the validation accuracy is not improving. This helps to save training time and prevent the declination of model performance.

To summarise the hyperparameters, learning rate of 0.00004 with AdamW optimiser make certain that the training of model is smooth and stable along with good generalisation. The cosine annealing schedulers make dynamic adjustments to avoid facing the local minima. The cross entropy loss ensure the model achieve most effective solution. As for early stopping, it helps to save training time and prevent overfitting from happening. Table 4.1 shows the summary of hyperparameters used in these experiments.

Table 4.1: Hyperparameters

Hyperparameters	Configuration
Batch size	64
Loss function	CrossEntropyLoss
Optimiser	AdamW
Learning rate	0.00004
Weight decay	0.0001
Scheduler	CosineAnnealingWarmRestarts
Early stopping	10

4.6 Model Validation

In this phase, the models are validated where it changes the mode to evaluation by disabling dropout and using the frozen batch norm stats. It also disables

computation of gradient for faster and memory-efficient validation techniques. Later, it performs inference and calculates the loss without updating the gradients. The validation accuracy is also tracked for the model checkpointing process. The model checkpoint is saved if the current validation accuracy has better value compared to previous epoch. It will reset the early stopping counter to zero when it saves the checkpoint. The training will terminate early if the validation accuracy does not improve for 10 consecutive epochs to prevent overfitting and save computation time. The time tracking function is also implemented to measure the time taken and remaining time to completely train all epochs. It is useful to monitor the progress especially it takes long training time.

4.7 Experiments

In order to assess the performance of pre-trained ViT, three different experiments are carried out for 90° viewing angles with all the three conditions such as normal walking, wearing a coat and carrying a bag. It involves 124 subjects and ten sequences. This model is evaluated in order to assess how GEI affects the accuracy of pre-trained ViT model with different conditions, particularly with a fixed viewing perspective that is 90°.

4.7.1 Experiment 1

This experiment is used to assess the performance of training and testing only the normal walking conditions. The filename for the dataset generated is GEI_Output_nm_organised which contains only the six sequences of normal walking which is nm-01, nm-02, nm-03, nm-04¸ nm-05¸ nm-06. This dataset is split randomly where the train set is 80% and test set is 20%. A validation set of 15% is also created based on the training set. The completion time of training for this experiment is 2 hours and 34 minutes where the early stopping is triggered at epoch 51 with a training accuracy of 100% and validation accuracy of 97.75%.

The test loss is 1.8008 which shows that the model makes few mistakes on test set. This indicates that the significance of absolute value depends on the specific loss function used. Lower test loss represents better performance of model. The model

predicts the correct output for 97.32% test samples which is high accuracy and shows that the overall performance is strong. The F1 score of 97.09% means that the precision and recall are balanced making it suitable for detecting true positives and minimising false positives or negatives. As for a precision of 97.99%, the model made almost all positive predictions correct. The recall of 97.32% shows that the identification of positive cases is successful. The result of final evaluation on the test set is summarised in Table 4.2 below.

Table 4.2: Experiment 1 Results

Metrics	Results
Test loss	1.8008
Accuracy	97.32%
F1 Score	97.09%
Precision	97.99%
Recall	97.32%

Figure 4.4 shows the graph for training and validation loss which are plotted to guide the process of optimisation where the loss represents the differences between predicted output of model and actual target value. In this graph, the x-axis is the number of epochs which is 50 and the y-axis is the loss value in which lower value indicating better performance. The blue line is the train loss that is calculated on training dataset after one epoch whereas the orange line is the validation loss that is calculated on training dataset after one epoch.

Both training and validation losses start at around 5 in which the value is quite high. This indicates that the initial performance on both datasets is poor. Later, both losses decrease continuously as the quantity of epochs increases. The model can be seen that it is improving through learning process. There is no sudden increase in validation loss and decrease in training loss which proves that the model is not overfit and generalises well to validation set.



Figure 4.4: Training and Validation Loss

Figure 4.5 shows the graph of training and validation accuracy are also plotted to observe the model's performance as training progresses. The x-axis is the number of epochs which is 50 and the y-axis is the accuracy such as the correct predictions. The blue line is the train accuracy that is calculated on training dataset after one epoch whereas the orange line is the validation accuracy that is calculated on training dataset after one epoch.

At early epochs which is 0 to 10, both accuracies start near 0 which is low as the model just start to learn. The training accuracy increases as the model learn patterns in training data in a fast way. The validation accuracy increases irregularly indicates that the model starts to generalise and is not stable yet. In the middle of epochs such as 10 to 30, training accuracy increases till it reaches 1.0 which is 100% whereas the validation accuracy increases slowly and irregularly with some drops and peaks which shows that the model is learning. Later, at epoch 30 to 50, the training accuracy remains near 1.0 and the validation accuracy is close to 1.0. The general trend is moving upwards which means that the model generalise properly as the training progresses.

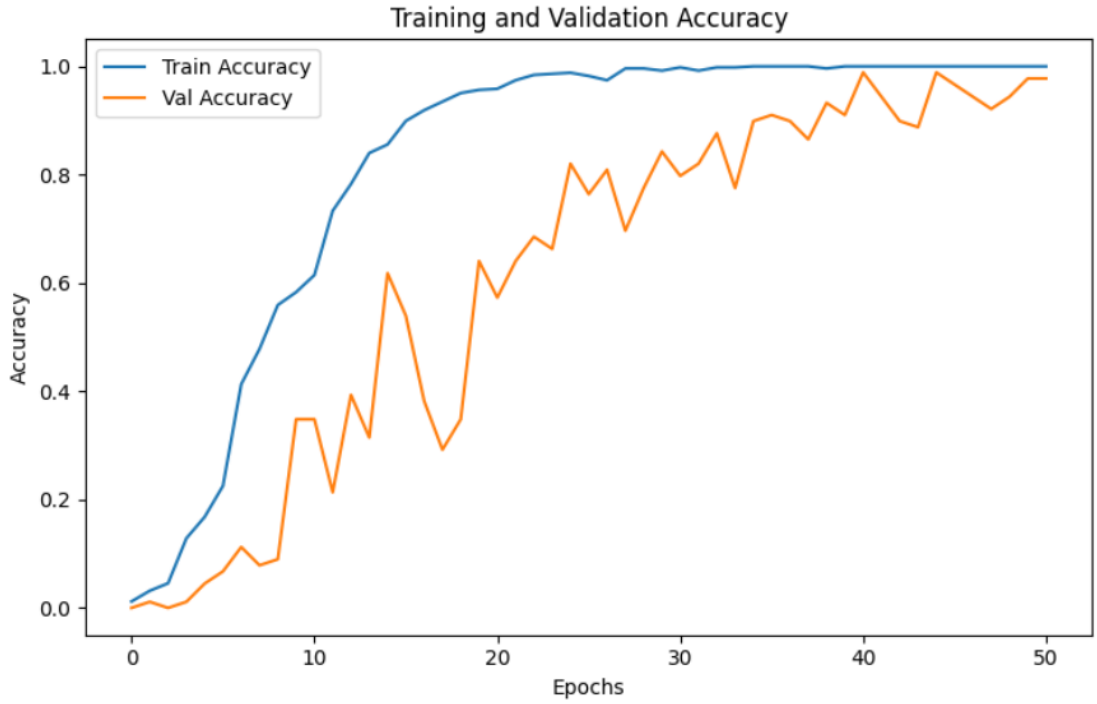


Figure 4.5: Training and Validation Accuracy

The confusion matrix in Figure 4.6 visualises and summarise the results of predictions against actual known values. The x-axis represents the class labels that is predicted whereas the y-axis represents actual class labels of data samples. Most of the colour intensity is at the diagonal which means that most predictions made are correct where the predicted class matches the actual class. There are a few light blue spots along the diagonal which means there is some misclassifications and it most probably involves 1 to 2 samples per class. As for the colour bar on the right, it ranges from 0 to 4 which means the highest misclassified number of samples between two classes are 4. It can be seen that most cells are near 0 which proves that it achieves high accuracy.

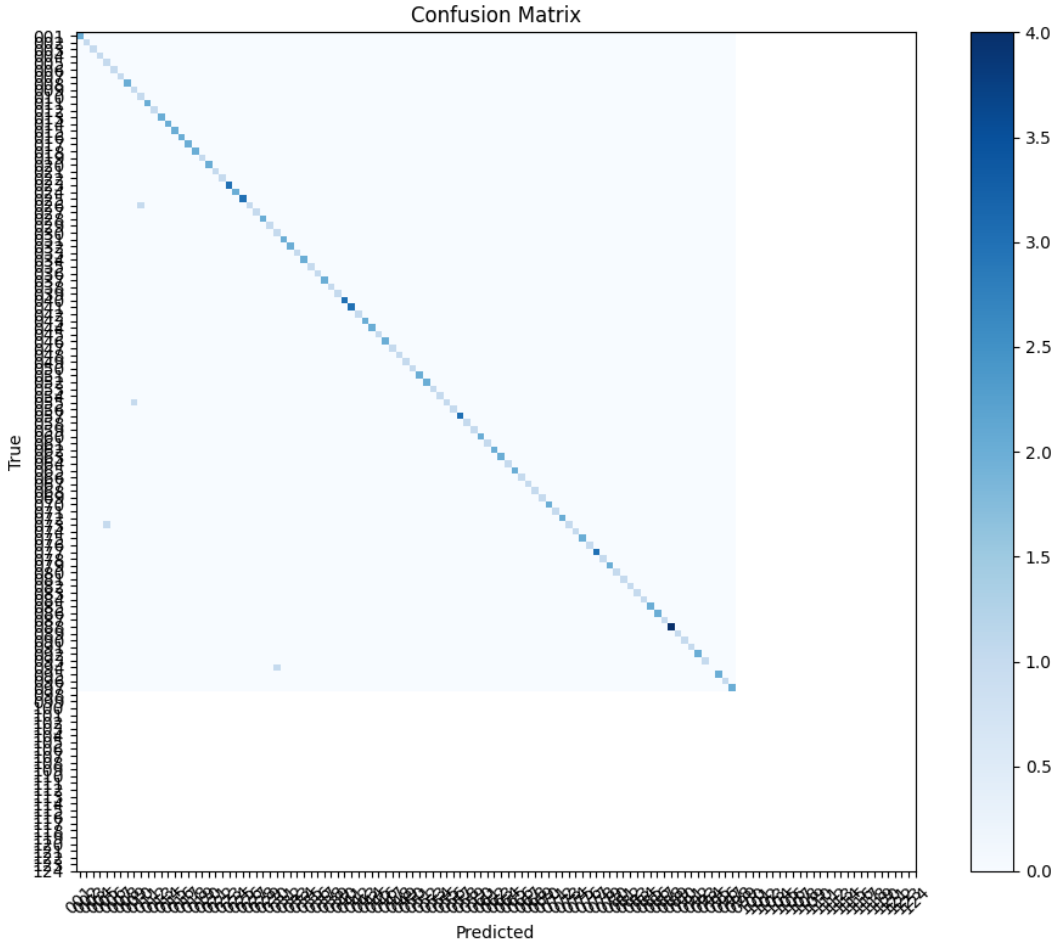


Figure 4.6: Confusion Matrix

4.7.2 Experiment 2

This experiment is used to assess the performance of training the combination of normal walking, wearing a coat and carrying a bag condition while testing each of the condition. The filename for the training dataset generated is GEI_Output_nm_cl_bg which contains four sequences of normal walking, one sequence of wearing a coat and one sequence of carrying a bag which is nm-01, nm-02, nm-03, nm-04, cl-01 and bg-01. As for the testing dataset, it has three datasets in which the first dataset is GEI_Output_test_nm for testing the normal walking conditions which has two sequences such as nm-05 and nm-06. The second dataset for testing is the wearing a coat condition which is GEI_Output_test_cl that has one sequence of cl-02 and third dataset for testing is the carrying a bag condition which is GEI_Output_test_bg that has one sequence that is bg-02.

4.7.2.1 Testing with normal walking condition

The first testing dataset is GEI_Output_test_nm which has a completion training time of 3 hours and 40 minutes where the early stopping is triggered at epoch 56 with a training accuracy of 100% and validation accuracy of 86.49%. The test loss is 1.8575 which shows that only few mistakes on test set are made by the model. This indicates that the significance of absolute value depends on the specific loss function used. Lower test loss represents better performance of model. The model predicts the correct output for 97.18% test samples which is high accuracy and shows that the excellent overall performance. The F1 score of 97.02% means that the precision and recall are balanced which makes it suitable for detecting true positives and minimising false positives or negatives. As for a precision of 97.98%, the model almost made all positive predictions correct. The recall of 97.18% shows that the detection of positive cases is successful. The result of final evaluation on the test set is shown in Table 4.3 below.

Table 4.3: Normal walking Results

Metrics	Results
Test loss	1.8575
Accuracy	97.18%
F1 Score	97.02%
Precision	97.98%
Recall	97.18%

Figure 4.7 shows the graph for training and validation loss which are plotted to guide the process of optimisation where the loss represents the differences between predicted output of model and actual target value. In this graph, the x-axis is the number of epochs which is about 55 and the y-axis is the loss value in which lower value indicating better performance as the prediction is closer to actual values. The blue line is the train loss that is calculated on training dataset after one epoch whereas the orange line is the validation loss that is calculated on training dataset after one epoch.

Initially, both training and validation losses start at around 5 in which the value is quite high. This indicates that the model starts with small knowledge and makes large errors. Later, both losses decrease as the number of epochs increases. This shows

that the model is improving through learning process by learning the patterns in a fast way. During the middle epoch 10 to 30, both losses decreases but the training loss decreases more sharply as it fits the training data. At epoch 30 to 55, the training loss reaches the value of approximately 1 as it continues to decline. The validation loss reaches around 2 to 2.5 which is its final loss. To conclude, both losses decrease initially which shows effective learning.

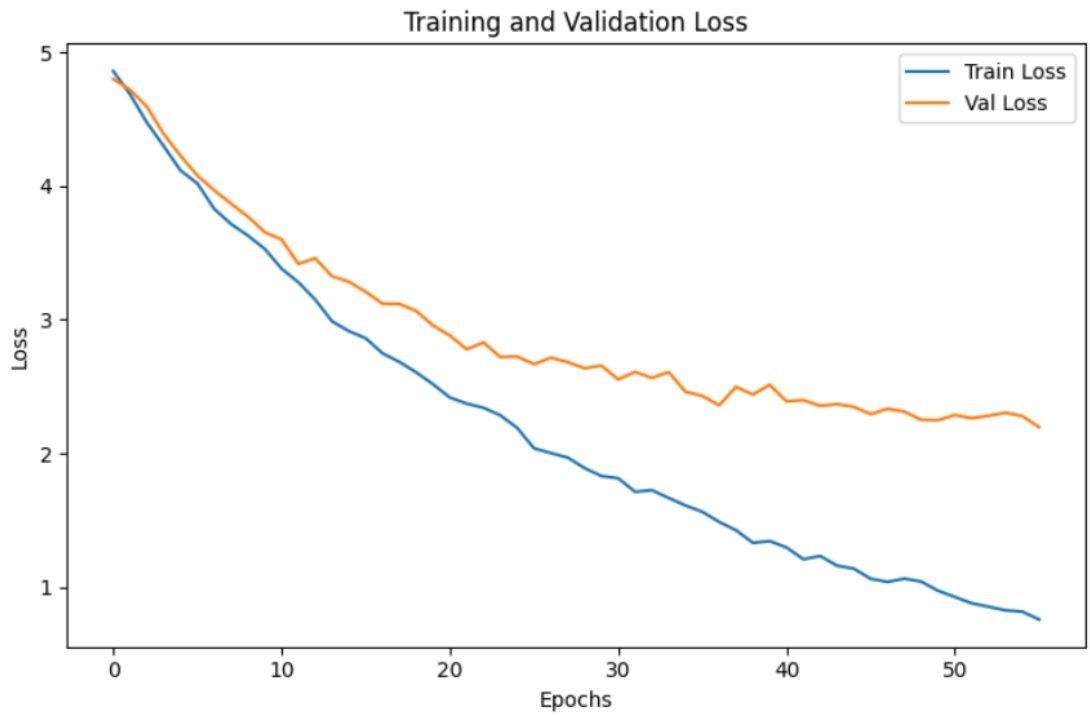


Figure 4.7: Training and Validation Loss

Figure 4.8 shows the graph of training and validation accuracy are also plotted to observe the model's performance as training progresses. The x-axis is the number of epochs which is around 0 to 55 and the y-axis is the accuracy such as the correct predictions. The blue line is the train accuracy that is calculated on training dataset after one epoch whereas the orange line is the validation accuracy that is calculated on training dataset after one epoch.

At early epochs which is 0 to 10, both lines start close to 0 which is low as the model has no knowledge and just start to learn. The training accuracy and validation accuracy increases as the model learn patterns in training data quickly. In the middle of epochs such as 10 to 25, training accuracy increases till it reaches 0.9 and above

which is 90% whereas the validation accuracy increases slowly and reach roughly 0.8 which is 80%. Later, at epoch 25 to 55, the training accuracy remains near 1.0 and the validation accuracy stabilise around 80 to 85%. The general trend is moving upwards which indicates that the model generalise well but include some instability as the training progresses.

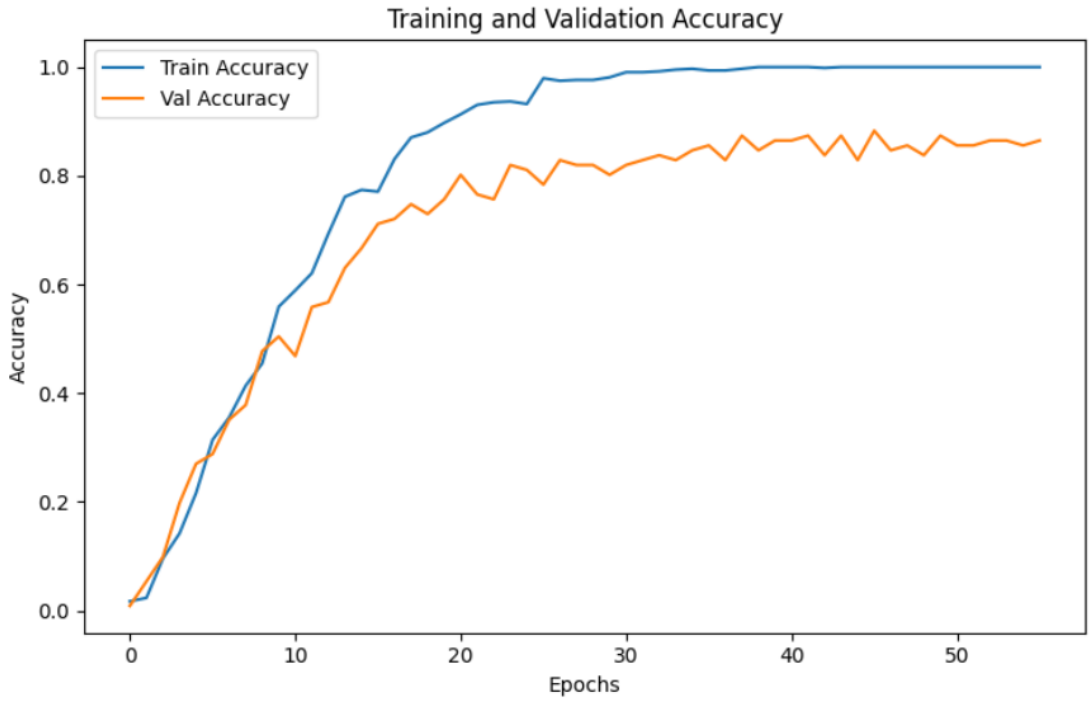


Figure 4.8: Training and Validation Accuracy

The confusion matrix in Figure 4.9 visualise and summarise the results of predictions against actual values. The x-axis represents the class labels that is predicted whereas the y-axis represents actual class labels of data samples. As for the colour scale, it ranges from 0 to 2 where 2 is the dark blue which represents higher number of samples for that cell. The darkness of diagonal is consistent which is the dark blue squares that appear the most. This shows that the predictions match the actual label and the model performs well across all classes. There are only a few light blue squares appear which shows that the number of wrongly predicted class are minimum. To conclude, the model is robust and it accurately predicts the correct class for most of the samples.

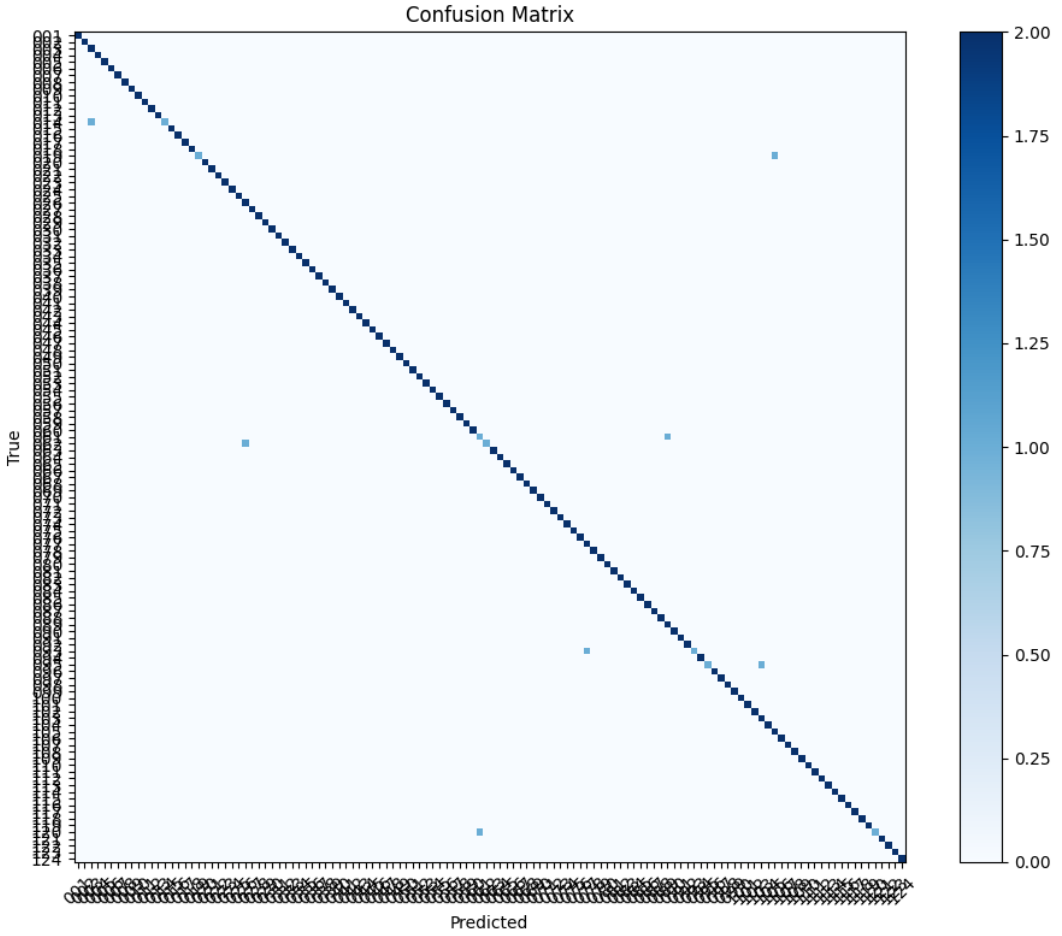


Figure 4.9: Confusion Matrix

4.7.2.2 Testing with wearing a coat condition

The second testing dataset is GEI_Output_test_cl in which the completion time of training is 4 hours and 22 minutes where the early stopping is triggered at epoch 66 with a training accuracy of 100% and validation accuracy of 88.29%. The test loss is 2.4775 which shows that the predictions are quite close to actual labels with the need of some improvements. This indicates that the significance of absolute value depends on the type of loss function used. Lower test loss represents better performance of model. The model predicts the correct output for 92.74% test samples which is high accuracy and shows that the model is reliable. The F1 score of 90.59% means that the model has precision and recall that are balanced which makes it suitable for detecting true positives and minimising false positives or negatives. As for a precision of 89.65%, the model made all positive predictions correct most of the time but there are still some false positives. The recall of 92.74% shows that it successfully identifies actual cases

with little false negatives. The result of final evaluation on the test set is summarised in Table 4.4 below.

Table 4.4: Wearing a coat Results

Metrics	Results
Test loss	2.4775
Accuracy	92.74%
F1 Score	90.59%
Precision	89.65%
Recall	92.74%

Figure 4.10 shows the graph for training and validation loss which are plotted to compare the loss across epochs based on training dataset and validation dataset. In this graph, the x-axis is the number of epochs which is over 60 and the y-axis is the loss value in which lower value means the prediction is closer to actual values which shows better performance. The blue line is the train loss that is based on training dataset after one epoch whereas the orange line is the validation loss that is based on training dataset after one epoch.

Initially, both training and validation losses start at a value that is quite high which is around 5. This indicates that the model makes large errors as it starts with few knowledge. Later, both losses decrease quickly as the number of epochs increases. This shows that the model is improving by learning the basic patterns of data rapidly. During the middle epoch 10 to 30, both losses continue to decrease but the training loss decreases more as it fits the training data better. At epochs 30 to 65, the training loss reaches the value below 1 as it continues to decline. The validation loss stops at around 2 to 2.5 and has small fluctuations which shows that the improvement of model has slowed. To conclude, it has a consistent decreasing which is an effective learning curve.

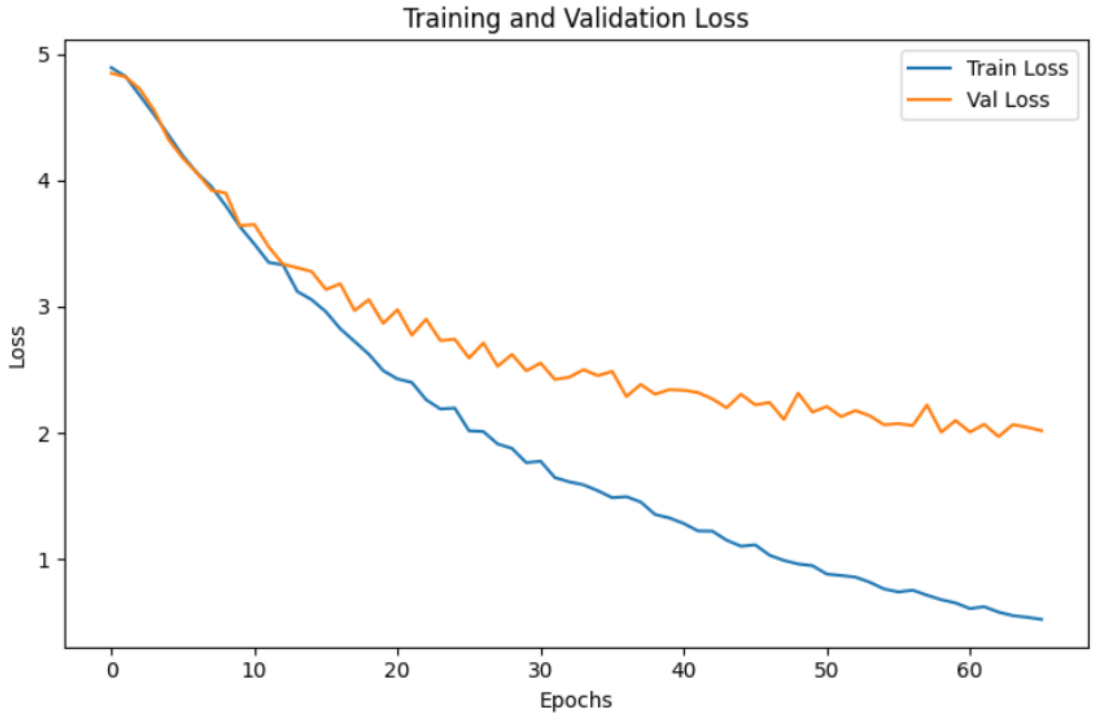


Figure 4.10: Training and Validation Loss

Figure 4.11 shows the graph of training and validation accuracy are also plotted to observe the performance of model as training progresses. The x-axis is the number of epochs which is around 0 to 65 and the y-axis is the accuracy such as the correct predictions. The blue line is the train accuracy that is calculated on training dataset after one epoch whereas the orange line is the validation accuracy that is calculated on training dataset after one epoch.

At early epochs which is 0 to 10, both lines start close to 0 which is low as the model has no knowledge and make mostly incorrect predictions. The training accuracy and validation accuracy increases as the model learn patterns in training data quickly. In the middle of epochs such as 10 to 30, training accuracy increases till it reaches 1.0 which is 100% whereas the validation accuracy increases slowly and reach around 0.8 which is 80%. Later, at epoch 30 to 65, the training accuracy remains at the top and the validation accuracy remain around 80 to 85% which indicates that the generalisation is stable but not perfect.

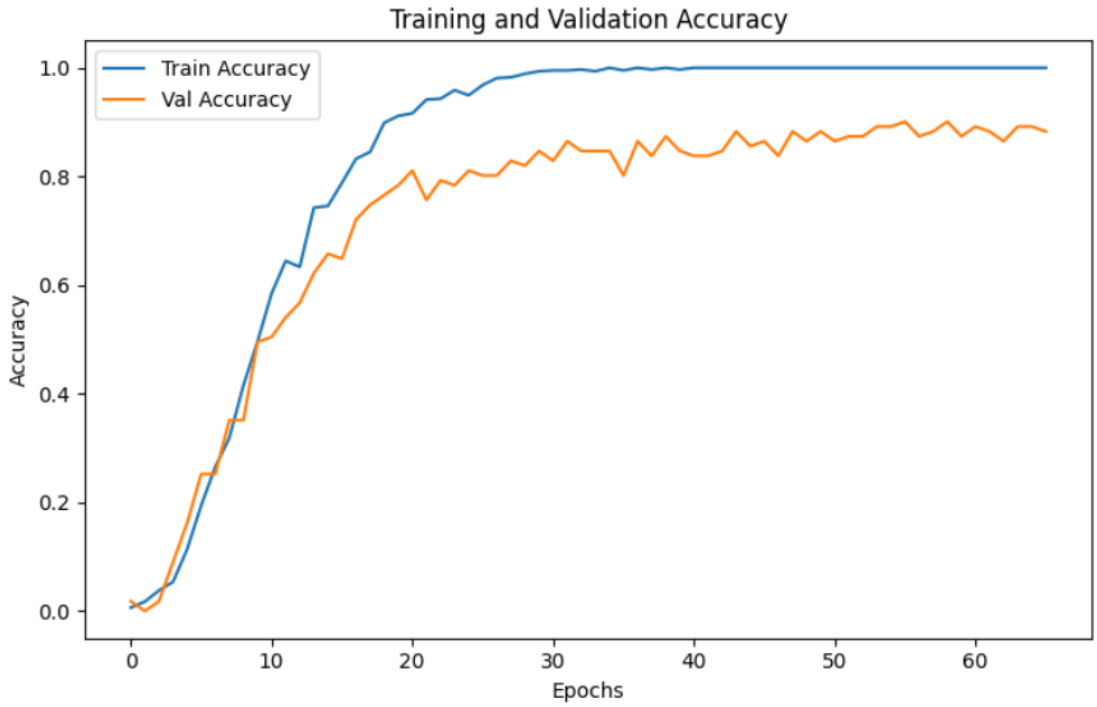


Figure 4.11: Training and Validation Accuracy

The confusion matrix in Figure 4.12 visualise and summarise the performance of model across the classes. The x-axis represents the class labels that is predicted whereas the y-axis represents actual class labels of data samples. The colour scale ranges from 0 to 1 where 1 is the dark blue. This dark blue represents a greater number of samples were correctly predicted as the class. The main diagonal is consistent which is the dark blue squares that appear the most. This shows that the predictions match the actual label and the model performs well across all classes. There are only a few light blue squares appear which shows that the number of wrongly predicted class are rare misclassification. To conclude, the performance of model is excellent as it has consistent high accuracy.

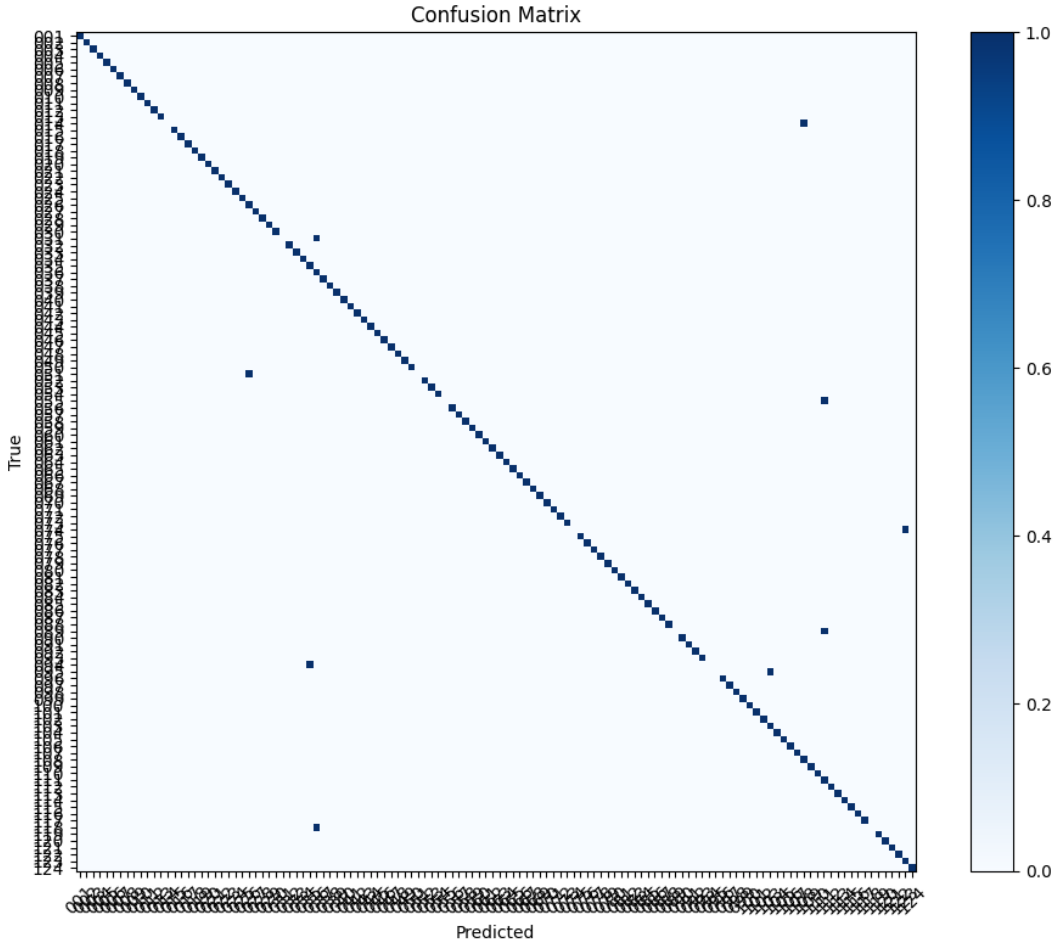


Figure 4.12: Confusion Matrix

4.7.2.3 Testing with carrying a bag condition

The third testing dataset is GEI_Output_test_bg in which the completion time of training is 3 hours and 08 minutes where the early stopping is triggered at epoch 50 with a training accuracy of 100% and validation accuracy of 89.19%. The test loss is 2.0993 where the predictions are near to actual labels with some improvements needed. This indicates that the significance of absolute value depends on the type of loss function used where lower test loss represents better performance of model. The model predicts the correct output for 94.35% which is high accuracy and shows that the model is good. The F1 score of 92.47% means that the model balanced precision and recall which makes it suitable for detecting true positives and minimising false positives or negatives. As for a precision of 91.53%, the model correctly predicts positive predictions most of the time. The recall of 94.35% shows that it successfully identifies actual cases with a few false negatives. The result of final evaluation on the test set is summarised in Table 4.5 below.

Table 4.5: Carrying a bag Results

Metrics	Results
Test loss	2.0993
Accuracy	94.35%
F1 Score	92.47%
Precision	91.53%
Recall	94.35%

Figure 4.13 shows the graph for training and validation loss which are plotted to view the loss across epochs based on the progression of training dataset and validation dataset. In this graph, the x-axis is the number of epochs which is 50 and the y-axis is the loss value in which lower value means the prediction is closer to actual values which indicates better performance. The train loss is based on training dataset after one epoch which is represented using blue line whereas the orange line is representing validation loss.

Initially, both training and validation losses decreases together which indicates that the model has learnt and improve on both datasets. During the middle epoch 10 to 20, both losses continue to decrease but the training loss reduces faster and validation loss decreases at slower rate. The model is fitting at training data as compared to validation data. At epochs 20 to 50, the training line reaches the value zero while the validation loss fluctuates a little. To conclude, the model learns during the initial stage as both losses decrease. This led to no evidence of underfitting as the training loss are quite low.

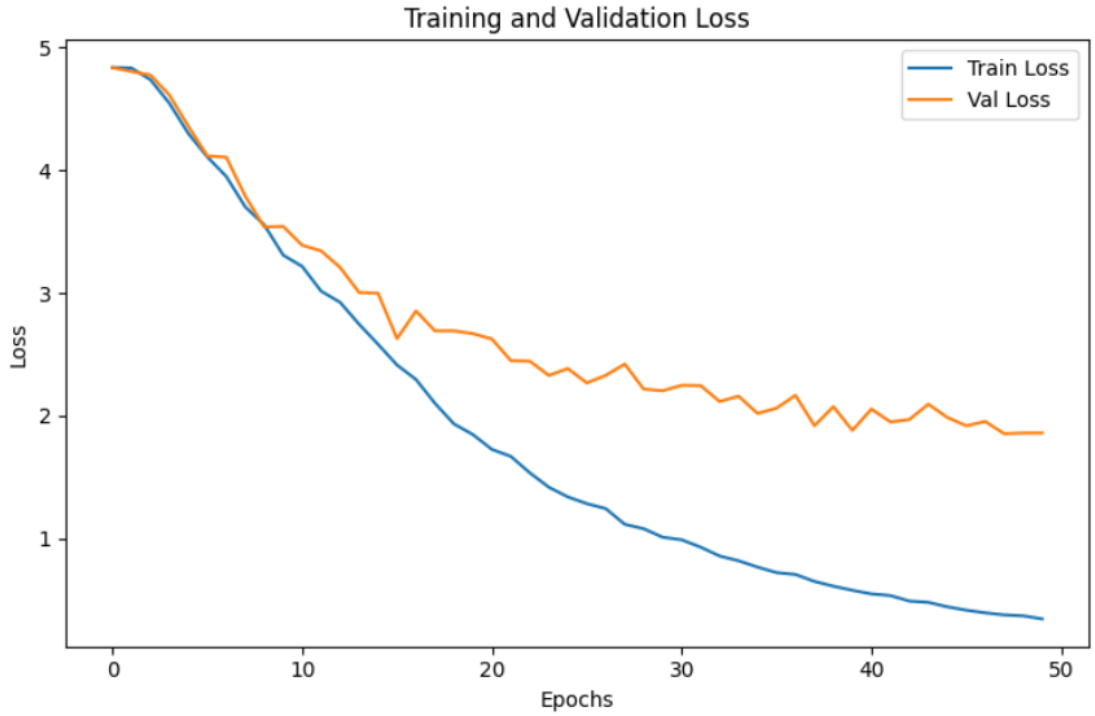


Figure 4.13: Training and Validation Loss

Figure 4.14 shows the graph of training and validation accuracy are also plotted to observe the accuracy evolving during training progresses. The x-axis is the number of epochs which is 0 to 50 and the y-axis is the accuracy such as the correct predictions. The blue line is the train accuracy that is calculated on training dataset after one epoch whereas the orange line is the validation accuracy that is calculated on training dataset after one epoch.

At early epochs of 0 to 10, both training and validation accuracy start near 0 and increases quickly through the steep increase. The model learn general features as the two lines are very close. In the middle phase where the number of epochs are 10 to 20, training accuracy increases till it reaches 1.0 which is 100% whereas the validation accuracy grows at slower rate and show irregular shift where it reaches around 0.8 which is 80%. Later, at epoch 20 to 50, the training accuracy remains close to 1.0 and the validation accuracy is around 85% which indicates that it has some fluctuations and does not improve. To conclude, the model learns and fits the training data well and at faster rate.

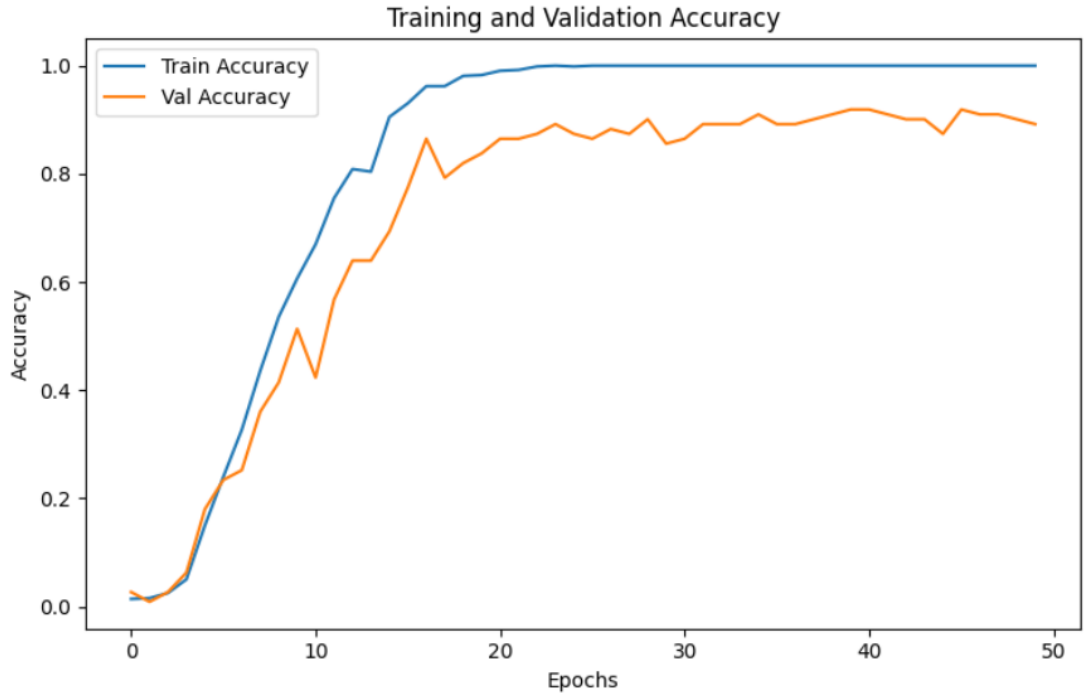


Figure 4.14: Training and Validation Accuracy

The confusion matrix in Figure 4.15 visualise and summarise the performance of model across the classes. The x-axis represents the class labels that is predicted whereas the y-axis represents actual class labels of data samples. The colour scale ranges from 0 to 1 where 1 is the dark blue. This dark blue represents a greater number of correct predictions which matches the actual label. The main diagonal is consistent as the dark blue squares appear the most. This shows that the predictions match the actual label and the model performs well across all classes. There are only a few squares appear outside of the diagonal which shows that it is the number of wrongly predicted class that are known as misclassification. To conclude, the performance of model is excellent as it has few misclassifications and it handle the complexity well.

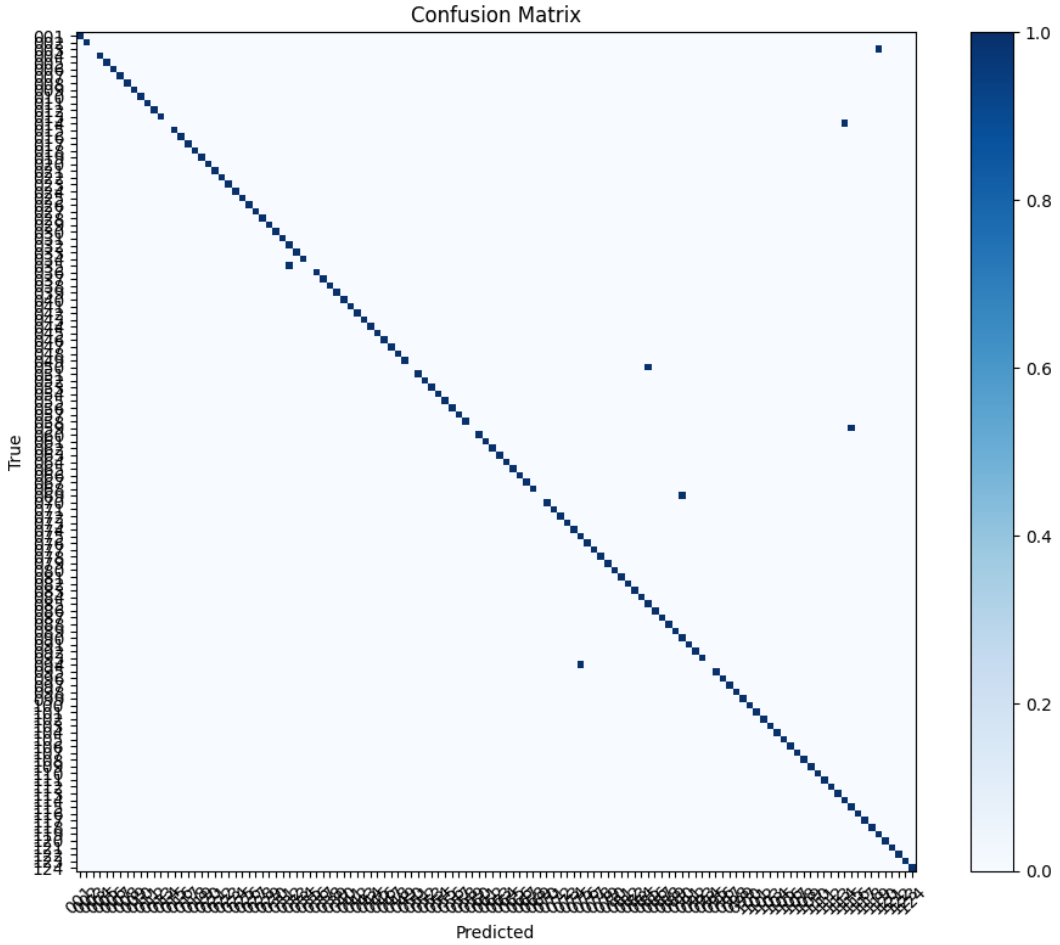


Figure 4.15: Confusion Matrix

4.7.3 Experiment 3

This experiment is used to assess the performance of training and testing combination of normal walking, wearing a coat and carrying a bag conditions. The filename for the dataset generated is GEI_Output_nm_cl_bg_all which contains all the six sequences of normal walking which is nm-01, nm-02, nm-03, nm-04¸ nm-05¸ nm-06. It also has 2 sequences of wearing a coat which is cl-01, cl-02 and 2 sequences of carrying a bag which is bg-01, bg-02. This dataset is split randomly where the train set is 80% and test set is 20%. A validation set of 15% is also created based on the training set. The completion time of training for this experiment is 6 hours and 04 minutes where the early stopping is triggered at epoch 70 with a training accuracy of 100% and validation accuracy of 97.30%.

The test loss is 1.4507 which shows that there are only few mistakes made by the model on test set. This indicates that the specific loss function used does affect the significance of absolute value. Lower test loss represents better performance of model. The model predicts the correct output for 97.58% test samples which shows that the overall performance is strong as it has high accuracy. The F1 score of 97.75% means that the precision and recall are balanced which makes it suitable for detecting true positives and minimising false positives or negatives. As for a precision of 98.59%, the model made almost all positive predictions correct. The recall of 97.58% shows that the identification of positive cases is successful. The result of final evaluation on the test set is summarised in Table 4.6 below.

Table 4.6: Experiment 3 Results

Metrics	Results
Test loss	1.4507
Accuracy	97.58%
F1 Score	97.75%
Precision	98.59%
Recall	97.58%

Figure 4.16 shows the graph plotted for training and validation loss to view the loss across epochs based on the progression of training dataset and validation dataset. In this graph, the x-axis is the number of epochs which is 70 and the y-axis is the loss value in which lower loss value means the prediction is closer to actual values which indicates better model performance. The blue line represents the train loss whereas the orange line represents the validation loss that is based on training dataset after one epoch.

Both training and validation loss decreases together from the starting value which is around 5. This shows that the learning process has help to improve the predictions on both datasets. Both losses continue to decrease during the middle epoch 10 to 40 where the training loss declines smoothly and validation loss decreases at slower rate with some fluctuations. The model is fitting at training data as compared to validation data. At epochs 40 to 70, the training loss decreases till it reaches the value zero while the validation loss continues to decrease consistently. To conclude, the model learns during the initial stage as both losses decrease.

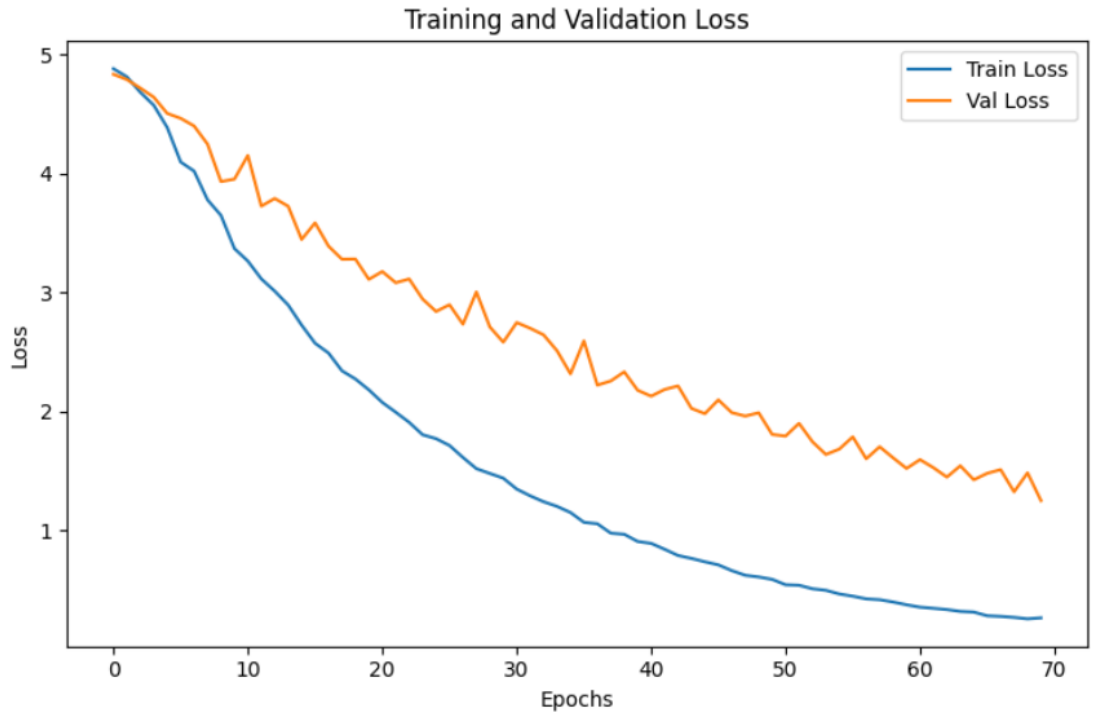


Figure 4.16: Training and Validation Loss

Figure 4.17 shows the graph of training and validation accuracy plotted to observe the accuracy evolving during training progresses. The x-axis is the number of epochs which is 0 to 70 and the y-axis is the accuracy such as the correct predictions. The blue line is the train accuracy that is calculated on training dataset after one epoch whereas the orange line is the validation accuracy that is calculated on training dataset after one epoch.

At early epochs of 0 to 10, both training and validation accuracy start near 0 which shows that it makes mostly wrong predictions. Then, it increases quickly through the learning process of useful features. In the middle phase where the number of epochs is 10 to 25, training accuracy increases till it reaches 1.0 which is 100% whereas the validation accuracy rises at slower rate and fluctuates. Later, at epoch 25 to 70, the training accuracy remains close to 1.0 and the validation accuracy is improving to around 90%. The model learns rapidly and achieves an almost perfect accuracy on training data.

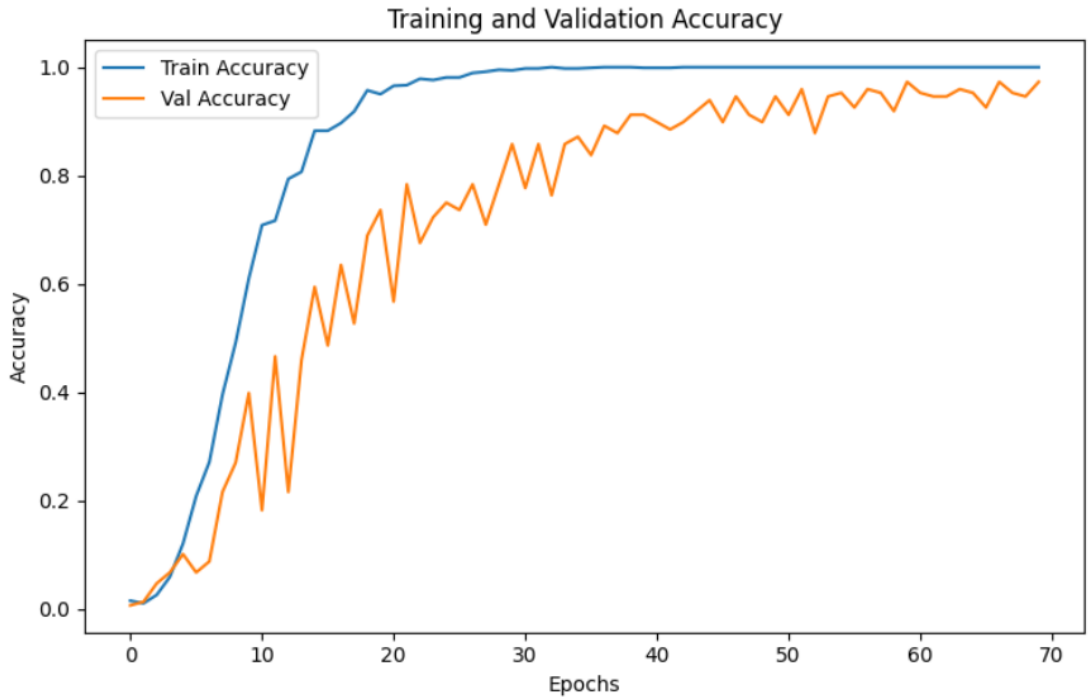


Figure 4.17: Training and Validation Accuracy

The confusion matrix in Figure 4.18 visualise and summarise the performance of model across the classes. The x-axis represents the class labels that is predicted whereas the y-axis represents actual class labels of data samples. The colour bar on the right ranges from 0 to 6 where 6 is the dark blue that represents higher counts. The main diagonal is consistent as it can be seen clearly which represents a greater number of correct predictions which matches the actual label. This shows that the predictions match the actual label and the model performs well across all classes. There are only a few squares appear outside of the diagonal which shows that there are only a few wrongly predicted classes that are known as misclassification. To conclude, the performance of model is good as it shows a strong diagonal and can handle the complexity well.

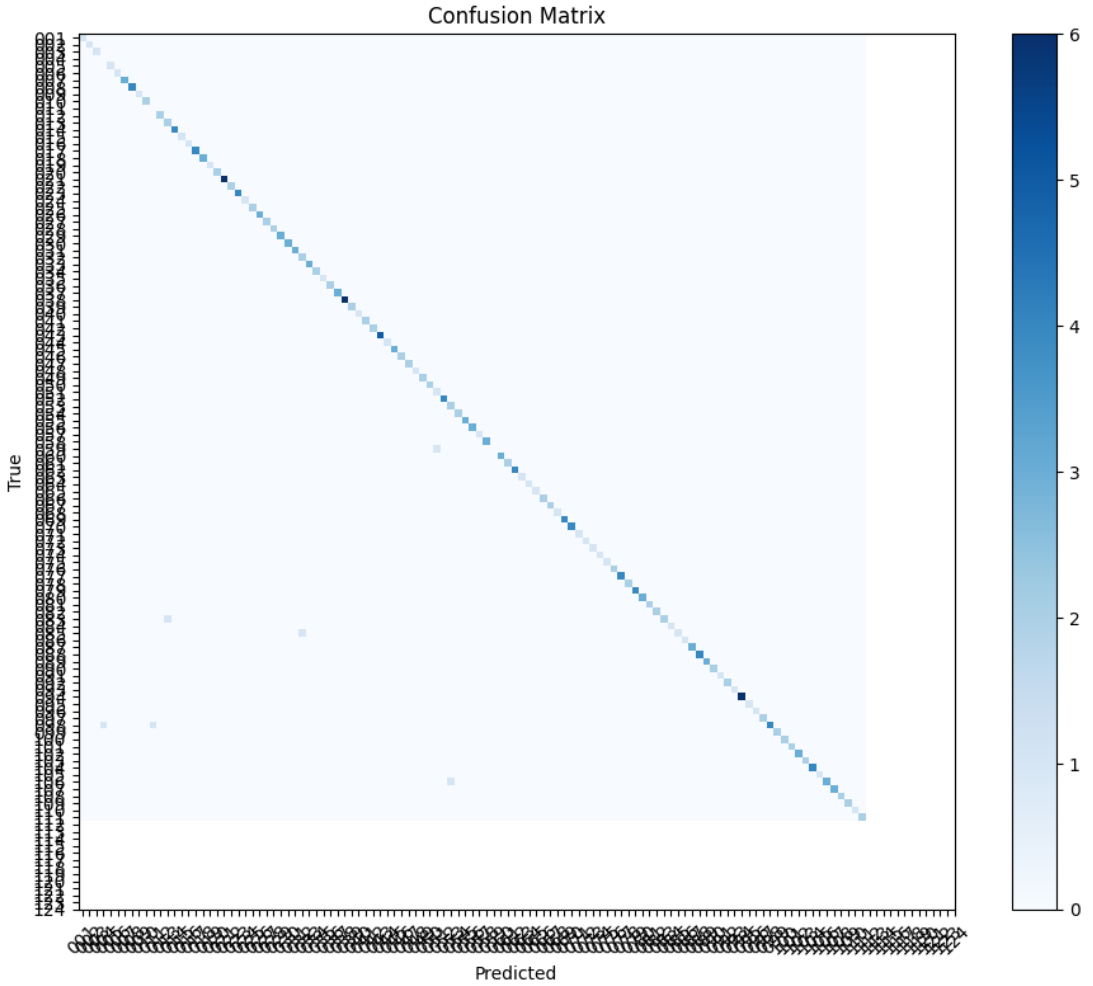


Figure 4.18: Confusion Matrix

4.8 Comparison with Existing Methods

Several existing models are used to compare the performance of this pre-trained ViT model in experiments that are done in this project. To compare the performance of model in Experiment 1, GaitSlice developed by H. Li et al. (2022) which is a CNN-based model that involves combinations of parallel RFAMs with the inter-related slice features from SED achieves 77.1%. This shows that this model achieves 97.32% which has better performance as compared to the GaitSlice.

As for the performance of model in Experiment 2 which is tested with normal walking condition, an existing CNN model which is an 8-layer architecture demonstrated by Sheth et al. (2023) achieves 95.45% as compared to this model achieving 97.18%.

In Experiment 2 with the testing of wearing a coat condition, an existing basic CNN model which has 10 layers proposed by Junaid & Ari (2022) achieves 92.4% as compared to this model which achieves 92.74%.

Another comparison in Experiment 2 is testing with carrying a bag condition in which Junaid & Ari (2022) created this 10-layer CNN model and is used to compare with this model. It is shown that the 10-layer CNN model achieves only 81.3% and this model achieves 94.35%.

Lastly, Experiment 3 is compared with a cubic SVM presented by Saleem et al. (2021) which is a machine learning that achieves only 89.8% and this model achieves 97.58%. Table 4.7 shows the comparison of the all the existing model and our pre-trained ViT model.

Table 4.7: Comparison between existing models and our pre-trained ViT

Model	Training and Testing Condition		Accuracy (%)
GaitSlice (H. Li et al., 2022)	Training	Normal	77.10
8-layer CNN model (Sheth et al., 2023)			-
10-layer CNN model (Junaid & Ari, 2022)	Testing	Normal	-
Cubic SVM (Saleem et al., 2021)			-
Ours			**97.32**
GaitSlice (H. Li et al., 2022)	Training	Normal, Wearing a coat and Carrying a bag	-
8-layer CNN model (Sheth et al., 2023)			95.45
10-layer CNN model (Junaid & Ari, 2022)	Testing	Normal	-
Cubic SVM (Saleem et al., 2021)			-
Ours			**97.18**
GaitSlice (H. Li et al., 2022)	Training	Normal, Wearing a coat and	-
8-layer CNN model (Sheth et al., 2023)			-

		Carrying a bag	
10-layer CNN model (Junaid & Ari, 2022)	Testing	Wearing a coat	92.40
Cubic SVM (Saleem et al., 2021)			-
Ours			**92.74**
GaitSlice (H. Li et al., 2022)	Training	Normal, Wearing a coat and Carrying a bag	-
8-layer CNN model (Sheth et al., 2023)			-
10-layer CNN model (Junaid & Ari, 2022)	Testing	Carrying a bag	81.30
Cubic SVM (Saleem et al., 2021)			-
Ours			**94.35**
GaitSlice (H. Li et al., 2022)	Training	Normal, Wearing a coat and Carrying a bag	-
8-layer CNN model (Sheth et al., 2023)			-
10-layer CNN model (Junaid & Ari, 2022)	Testing	Normal, Wearing a coat and Carrying a bag	-
Cubic SVM (Saleem et al., 2021)			89.80
Ours			**97.58**

To conclude, this model proves that the performance is good as it achieves high accuracy as compared to existing models. These results show that this model is suitable for recognising gait in different experimental conditions. It can generalise well in different situations which makes it an advancement in deep learning.

4.9 Summary

These experiments evaluate the performance of a pre-trained ViT for gait recognition system using CASIA-B dataset which has different walking conditions across eleven viewing angles. Several important steps such as preprocessing and data augmentation are done before training the data to improve the performance of model.

Three experiments with different training and testing conditions are conducted to evaluate the results of model. Experiment 1 assess the model performance in identifying normal walking sequences and achieving an accuracy of 97.32%. Experiment 2 has more complex setup where it is trained on combination of normal, coat and bag sequences while testing each condition separately. This experiment achieves excellent performance such as 97.18% for normal walking, 92.74% for wearing a coat and 94.35% for carrying a bag. As for Experiment 3, it is train and test on combination of sequences where it achieves 97.58%. These results highlight the strength of transformer-based architectures in experimenting different conditions.

CHAPTER 5

CONCLUSION

5.1 Conclusion

The first objective of this research is to study existing research on gait recognition using silhouette images. To achieve this objective, the study of literature review that focuses on both traditional machine learning and deep learning-based gait recognition techniques are conducted. This literature review includes the advantages of using silhouette images to preserve spatial and temporal walking patterns while maintaining the privacy of individuals. There are several models reviewed such as CNNs, RNNs, and many more to identify different gait representations. The way silhouette-based methods are applied in different conditions shows that it is effective and suitable for gait recognition.

Next, the second objective is to develop a deep learning model which can identify individuals based on their own gait patterns that is extracted from raw silhouette images. This objective is done through the development of model such as pre-trained ViT that is fine-tuned on CASIA-B dataset. The preprocessing phase is done to extract the GEI representations from raw silhouette sequences. It is then trained with a custom classification head as a modified ViT model. The model is built to capture both spatial and temporal gait patterns which proves that it can differentiate between individuals based on their own walking patterns in different conditions and viewing angles.

Lastly, the final objective is to evaluate the performance and accuracy of developed model. This is accomplished through the development of three experiments that has different training and testing conditions. The model has achieved high accuracy in all the experiments where it is tested across different conditions like normal walking, wearing a coat and carrying a bag. There are several performance metrics used such as accuracy, precision, recall, f1 score and confusion matrix to assess the generalisability of the model. The high accuracy was recorded which is 97.58%

indicating the model has the capability to identify individuals based on their gait patterns. This led to the success of developing this pre-trained Vision Transformer.

5.2 Future work

There are several future works to be done in order to improve the performance and applicability of the pre-trained vision transformer model in this project. This model may be applied in larger and more variations datasets such as GREW or OU-MVLP which involves outdoor environments, various lighting conditions and complex backgrounds. This may help to improve the ability of model to generalise under real-world conditions by evaluating the performance of model under different conditions.

Additionally, it can experiment with lightweight type of Vision Transformer or any suitable hybrid models that combine CNN and transformer-based which helps to reduce computational resources and increase the accuracy. This is important to ensure that the deployment of model is efficient in handling resources. Next, the training process can also include more data augmentation techniques which can make the model more adaptable to many variations. The future work aims to improve this gait recognition framework to be more effective and reliable.

REFERENCES

Asif, M., Tiwana, M. I., Khan, U. S., Ahmad, M. W., Qureshi, W. S., & Iqbal, J. (2022). Human gait recognition subject to different covariate factors in a multi-view environment. *Results in Engineering*, *15*. https://doi.org/10.1016/j.rineng.2022.100556

Chao, H., Wang, K., He, Y., Zhang, J., & Feng, J. (2022). GaitSet: Cross-View Gait Recognition Through Utilizing Gait As a Deep Set. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*(7), 3467–3478. https://doi.org/10.1109/TPAMI.2021.3057879

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2020). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. http://arxiv.org/abs/2010.11929

Filipi Gonçalves Dos Santos, C., Oliveira, D. D. S., Passos, L. A., Gonçalves Pires, R., Felipe Silva Santos, D., Pascotti Valem, L., Moreira, T. P., Santana, M. C. S., Roder, M., Paulo Papa, J., & Colombo, D. (2021). Gait Recognition Based on Deep Learning: A Survey. In *ACM Computing Surveys* (Vol. 55, Issue 2). Association for Computing Machinery. https://doi.org/10.1145/3490235

Gul, S., Malik, M. I., Khan, G. M., & Shafait, F. (2021). Multi-view gait recognition system using spatio-temporal features and deep learning. *Expert Systems with Applications*, *179*. https://doi.org/10.1016/j.eswa.2021.115057

Güner Şahan, P., Şahin, S., & Kaya Gülağız, F. (2024). A survey of appearance-based approaches for human gait recognition: techniques, challenges, and future directions. *Journal of Supercomputing*, *80*(13), 18392–18429. https://doi.org/10.1007/s11227-024-06172-z

Hanif, C. A., Mughal, M. A., Khan, M. A., Almujally, N. A., Kim, T., & Cha, J. H. (2024). Human Gait Recognition for Biometrics Application Based on Deep Learning Fusion Assisted Framework. *Computers, Materials and Continua*, *78*(1), 357–374. https://doi.org/10.32604/cmc.2023.043061

Içik, S. G., & Ekenel, H. K. (2021). Deep Convolutional Feature-based Gait Recognition Using Silhouettes and RGB Images. *Proceedings - 6th International Conference on Computer Science and Engineering, UBMK 2021*, 336–341. https://doi.org/10.1109/UBMK52708.2021.9559026

Junaid, I., & Ari, S. (2022). Gait Recognition under Different Covariate Conditions using Deep Learning Technique. *SPCOM 2022 - IEEE International Conference on Signal Processing and Communications*. https://doi.org/10.1109/SPCOM55316.2022.9840818

Kececi, A., Yildirak, A., Ozyazici, K., Ayluctarhan, G., Agbulut, O., & Zincir, I. (2020). Implementation of machine learning algorithms for gait recognition. *Engineering Science and Technology, an International Journal*, *23*(4), 931–937. https://doi.org/10.1016/j.jestch.2020.01.005

Khan, A., Khan, M. A., Javed, M. Y., Alhaisoni, M., Tariq, U., Kadry, S., Choi, J. I., & Nam, Y. (2022). Human gait recognition using deep learning and improved ant colony optimization. *Computers, Materials and Continua*, *70*(2), 2113–2130. https://doi.org/10.32604/cmc.2022.018270

Khan, M. H., Farid, M. S., & Grzegorzek, M. (2021). Vision-based approaches towards person identification using gait. In *Computer Science Review* (Vol. 42). Elsevier Ireland Ltd. https://doi.org/10.1016/j.cosrev.2021.100432

Kukreja, V., Kumar, D., & Kaur, A. (2021). Deep learning in Human Gait Recognition: An Overview. *2021 International Conference on Advance Computing and Innovative Technologies in Engineering, ICACITE 2021*, 9–13. https://doi.org/10.1109/ICACITE51222.2021.9404611

Kumar, M., Singh, N., Kumar, R., Goel, S., & Kumar, K. (2021). Gait recognition based on vision systems: A systematic survey. *Journal of Visual Communication and Image Representation*, *75*. https://doi.org/10.1016/j.jvcir.2021.103052

Li, H., Qiu, Y., Zhao, H., Zhan, J., Chen, R., Wei, T., & Huang, Z. (2022). GaitSlice: A gait recognition model based on spatio-temporal slice features. *Pattern Recognition*, *124*. https://doi.org/10.1016/j.patcog.2021.108453

Li, X., Makihara, Y., Xu, C., Yagi, Y., Yu, S., & Ren, M. (n.d.). *End-to-end Model-based Gait Recognition*.

Narayan, V., Awasthi, S., Fatima, N., Faiz, M., & Srivastava, S. (2023). Deep Learning Approaches for Human Gait Recognition: A Review. *2023 International Conference on Artificial Intelligence and Smart Communication, AISC 2023*, 763–768. https://doi.org/10.1109/AISC56616.2023.10085665

Nazari, F., Mohajer, N., Nahavandi, D., & Khosravi, A. (2022). Comparison of gait phase detection using traditional machine learning and deep learning techniques. *Conference Proceedings - IEEE International Conference on Systems, Man and Cybernetics*, *2022-October*, 403–408. https://doi.org/10.1109/SMC53654.2022.9945397

Parashar, A., Parashar, A., Ding, W., Shabaz, M., & Rida, I. (2023). Data preprocessing and feature selection techniques in gait recognition: A comparative study of machine learning and deep learning approaches. *Pattern Recognition Letters*, *172*, 65–73. https://doi.org/10.1016/j.patrec.2023.05.021

Peltekis, C., Alexandridis, K., & Dimitrakopoulos, G. (2024). Reusing Softmax Hardware Unit for GELU Computation in Transformers. *2024 IEEE 6th International Conference on AI Circuits and Systems, AICAS 2024 - Proceedings*, 159–163. https://doi.org/10.1109/AICAS59952.2024.10595882

Raihen, M. N., & Akter, S. (2024). Prediction modeling using deep learning for the classification of grape-type dried fruits. *International Journal of Mathematics and Computer in Engineering*, *2*(1), 1–12. https://doi.org/10.2478/ijmce-2024-0001

Saleem, F., Khan, M. A., Alhaisoni, M., Tariq, U., Armghan, A., Alenezi, F., Choi, J. I., & Kadry, S. (2021). Human gait recognition: A single stream optimal deep learning features fusion. *Sensors*, *21*(22). https://doi.org/10.3390/s21227584

Sayeed, M. S., Min, P. P., & Bari, M. A. (2022). Deep Learning Based Gait Recognition Using Convolutional Neural Network in the COVID-19 Pandemic. *Emerging Science Journal*, *6*(5), 1086–1099. https://doi.org/10.28991/ESJ-2022-06-05-012

Sepas-Moghaddam, A., & Etemad, A. (2023). Deep Gait Recognition: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(1), 264–284. https://doi.org/10.1109/TPAMI.2022.3151865

Sharif, M. I., Khan, M. A., Alqahtani, A., Nazir, M., Alsubai, S., Binbusayyis, A., & Damaševičius, R. (2022). Deep Learning and Kurtosis-Controlled, Entropy-Based Framework for Human Gait Recognition Using Video Sequences. *Electronics (Switzerland)*, *11*(3). https://doi.org/10.3390/electronics11030334

Sheth, A., Sharath, M., Reddy, S. C., & Sindhu, K. (2023). Gait Recognition Using Convolutional Neural Network. *International Journal of Online and Biomedical Engineering*, *19*(1), 107–118. https://doi.org/10.3991/ijoe.v19i01.33823

Slemenšek, J., Fister, I., Geršak, J., Bratina, B., van Midden, V. M., Pirtošek, Z., & Šafarič, R. (2023). Human Gait Activity Recognition Machine Learning Methods. *Sensors*, *23*(2). https://doi.org/10.3390/s23020745

Song, C., Yun, L., & Li, R. (2024). MFCF-Gait: Small Silhouette-Sensitive Gait Recognition Algorithm Based on Multi-Scale Feature Cross-Fusion. *Sensors*, *24*(17). https://doi.org/10.3390/s24175500

Yao, L., Kusakunniran, W., Wu, Q., Zhang, J., Tang, Z., & Yang, W. (2021). Robust gait recognition using hybrid descriptors based on Skeleton Gait Energy Image. *Pattern Recognition Letters*, *150*, 289–296. https://doi.org/10.1016/j.patrec.2019.05.012

Zhou, T., Niu, Y., Lu, H., Peng, C., Guo, Y., & Zhou, H. (2024). Vision transformer: To discover the “four secrets” of image patches. *Information Fusion*, *105*. https://doi.org/10.1016/j.inffus.2024.102248

Zhu, Z., Guo, X., Yang, T., Huang, J., Deng, J., Huang, G., Du, D., Lu, J., & Zhou, J. (n.d.). *Gait Recognition in the Wild: A Benchmark*. https://www.grew-benchmark.org/.

Zou, Q., Wang, Y., Wang, Q., Zhao, Y., & Li, Q. (2020). Deep Learning-Based Gait Recognition Using Smartphones in the Wild. *IEEE Transactions on Information Forensics and Security*, *15*, 3197–3212. https://doi.org/10.1109/TIFS.2020.2985628

APPENDICES

Appendix A: FYP Meeting Log



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 27/3/2025	**MEETING NO.: 1**
PROJECT ID: T86O641	
PROJECT TITLE : GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH	
TRIMESTER : Trimester Mar/Apr 2025	**SUPERVISOR : Pa Pa Min**
STUDENT ID & Name: 1211102837 Lysa Liew Wan Jin	**CO- SUPERVISOR :**

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Perform several times of training and testing on the pretrained Vision Transformer model to achieve good performance
2. WORK TO BE DONE Train and test the model with early stopping to avoid long training time
3. PROBLEMS ENCOUNTERED
4. COMMENTS

Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

NOTES:
1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)
**: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MULTIMEDIA UNIVERSITY

Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

MEETING DATE: 10/4/2025	**MEETING NO.: 2**
PROJECT ID: T86O641	
PROJECT TITLE : GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH	
TRIMESTER : Trimester Mar/Apr 2025	**SUPERVISOR : Pa Pa Min**
STUDENT ID & Name: 1211102837 Lysa Liew Wan Jin	**CO- SUPERVISOR :**

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Perform several times of training and testing on the pretrained Vision Transformer model to achieve good performance
2. WORK TO BE DONE Search for gait-related or motion-related Vision Transformer to load it as pretrained ViT
3. PROBLEMS ENCOUNTERED • Limited resource for gait-related Vision Transformer • The Vision Transformer found is trained using Keras library and it clashed with some part of the code as this project is trained using Pytorch library
4. COMMENTS

Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MULTIMEDIA UNIVERSITY

Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

MEETING DATE: 17/4/2025	**MEETING NO.: 3**
PROJECT ID: T86O641	
PROJECT TITLE : GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH	
TRIMESTER : Trimester Mar/Apr 2025	**SUPERVISOR : Pa Pa Min**
STUDENT ID & Name: 1211102837 Lysa Liew Wan Jin	**CO- SUPERVISOR :**

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.]
Perform several times of training and testing on the pretrained Vision Transformer model to achieve good performance

2. WORK TO BE DONE

- Train the same model with two different testing condition
- Train with normal condition and test with wearing clothes and carrying bag condition

3. PROBLEMS ENCOUNTERED
The performance of model is not satisfying

4. COMMENTS

Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)
**: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MULTIMEDIA UNIVERSITY

Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

MEETING DATE: 24/4/2025	**MEETING NO.: 4**
PROJECT ID: T86O641	
PROJECT TITLE : GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH	
TRIMESTER : Trimester Mar/Apr 2025	**SUPERVISOR : Pa Pa Min**
STUDENT ID & Name: 1211102837 Lysa Liew Wan Jin	**CO- SUPERVISOR :**

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Perform several training and testing on the pretrained Vision Transformer model to achieve good performance
2. WORK TO BE DONE • Train the same model with normal walking, wearing a coat and carrying a bag condition • Test the model with each condition
3. PROBLEMS ENCOUNTERED
4. COMMENTS

Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MULTIMEDIA UNIVERSITY

Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

MEETING DATE: 8/5/2025	**MEETING NO.: 5**
PROJECT ID: T86O641	
PROJECT TITLE : GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH	
TRIMESTER : Trimester Mar/Apr 2025	**SUPERVISOR : Pa Pa Min**
STUDENT ID & Name: 1211102837 Lysa Liew Wan Jin	**CO- SUPERVISOR :**

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.]
Perform several training and testing on the pretrained Vision Transformer model to achieve good performance

2. WORK TO BE DONE

- Train and test the same model with normal walking, wearing a coat and carrying a bag condition
- Write Chapter 3: Methodology and Chapter 4: Experimental Design and Analysis in Report

3. PROBLEMS ENCOUNTERED

4. COMMENTS

Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MULTIMEDIA UNIVERSITY

Faculty of Information Science and Technology (FIST)

Final Year Project Meeting Log

MEETING DATE: 29/5/2025	**MEETING NO.: 6**
PROJECT ID: T86O641	
PROJECT TITLE : GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH	
TRIMESTER : Trimester Mar/Apr 2025	**SUPERVISOR : Pa Pa Min**
STUDENT ID & Name: 1211102837 Lysa Liew Wan Jin	**CO- SUPERVISOR :**

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.]

- Train and test the same model with normal walking, wearing a coat and carrying a bag condition
- Complete Chapter 3: Methodology and Chapter 4: Experimental Design and Analysis in Report

2. WORK TO BE DONE

- Complete Chapter 4: Experimental Design and Analysis and Chapter 5: Conclusion in Report

3. PROBLEMS ENCOUNTERED

4. COMMENTS

Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

Appendix B: Technical Paper

Gait Recognition from Silhouette Images: A Deep Learning Approach

Lysa Wan Jin Liew
Faculty of Information Science and Technology, Multimedia University,
Melaka, Malaysia
1211102837@student.mmu.edu.my

Pa Pa Min
Faculty of Information Science and Technology
Multimedia University
Melaka, Malaysia
papamin@mmu.edu.my

***Abstract*—Gait recognition quickly emerges due to its ability to function remotely and without the need for the individual to actively engage. This project research into demonstrating the application of deep learning algorithms that use silhouette representations as input. The main objective is to overcome the problems of traditional gait recognition techniques that have low accuracy. This project uses multiview CASIA-B dataset that includes different walking conditions. The transformations of raw silhouettes into Gait Energy Image (GEI) which are two-dimensional representations of gait patterns are done. The primary outcome demonstrated the implementation of a pre-trained Vision Transformer (ViT) model for gait detection tasks as it showed that GEI works well in gathering spatiotemporal features. There are several experiments conducted under different walking conditions across the 90° viewing angle. The performance of model is evaluated using standard evaluation metrics and it proves that this model does not only achieve high accuracy in one walking condition, but across different combinations of conditions. The comparative analysis with existing gait recognition models further highlights the adaptability and suitability of this pre-trained ViT. Overall, this project creates an adaptable and effective system for recognising gait through the deployment of deep learning model.**

***Keywords*—Gait Recognition, Gait Energy Image (GEI), pre-trained ViT, CASIA-B**

I. INTRODUCTION

Gait recognition is a method of identifying an individual based on their physical and personality traits just by observing how they walk [1]. Each gait has its own set of characteristics where it identifies an individual without their assistance or physical contact with the system. This is a unique feature due to its unobtrusive design. The gait biometric method which involves figuring out individuals from their gait pattern has been widely acknowledged and used. According to [2], the capacity to function from a distance and effective in lower resolution films makes it more reliable.

Since the importance of silhouette images is not entirely understood, it is hard to determine how they contribute to improve the effectiveness and dependability of gait detection systems in comparison to normal photos. There is also a lack of advancement and limitation in the features found in traditional gait recognition systems as it is unable to reach high accuracy. The effectiveness, reliability, and interpretability of some gait recognition models are uncertain as some were created without conducting thorough evaluations.

The first objective of this research is to study existing research on gait recognition using silhouette images. To achieve this objective, the study of literature review focuses on both traditional machine learning and deep learning-based gait recognition techniques are conducted. The second objective is to develop a deep learning model which can identify individuals based on their own gait patterns that are extracted from raw silhouette images. This objective is done through the development of pre-trained ViT that is fine-tuned on CASIA-B dataset. The final objective is to evaluate the performance and accuracy of the developed model. This is accomplished through the development of three experiments that have different training and testing conditions.

II. LITERATURE REVIEW

A. Gait Recognition Approach

According to A. Khan et al. [3], it is demonstrated that there are two methods for recognising human gait which are model- based and model-free. The model-based method uses previous data to guide movement of humans. This method examines human behaviour by looking at upper and lower body components and movement of joints. The model-free method creates skeletons or postural generation which are representations of the human body. However, it is easier to build and takes less computing time to use the model-free technique.

Saleem et al. [4] researched that model-based and model-free methods are the two main subcategories of human gait recognition. Human gait is categorised using model-based methods such as movement and body shape models. A human body is described using an architectural model and defining gait by tracking various body components and movement of joints over time using basic mathematical concepts including stick-figure, interconnected pendulum and elliptical fitting approaches. The model-free approach which is easy to use because it requires less computing power uses the gait time to determine the attributes of the object silhouette.

According to Kumar et al. [5], there are two types of approaches which are model-free and model-based. Model-free method uses a binary silhouette to identify the gait of a person. A silhouette is a single-coloured, solid-shaped picture of someone or object. Typically, the inside of silhouette has fewer features and is coloured black. Therefore, applying background modelling methods is necessary for the extraction of silhouettes. A model-based approach uses information about a person's joints and other body elements. The quality of gait sequence needs to be high for this method. Therefore, a multi-camera arrangement is required to give the system high-quality gait sequences.

A. Khan et al. [3] demonstrated that the current gait identification methods may be grouped into two methods, which are model-based techniques and model-free techniques. Model-based methods use movement and structure of body models to create a gait characteristic. This method is independent of scale and vision. The skeletal models explain how to represent a person's torso, upper limbs, and lower body parts by calculating various measures on these areas like width and height. The motion analysis model which is applied for gait detection, focusses on the movement data of the body of

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

Appendix C: Checklist for FYP Final Report Submission



Faculty of Information Science and Technology (FIST)

Checklist for Final Report Submission

(To be filled in by Student)

STUDENT'S DETAILS

Project Code	**T86O641**
Name	LYSA LIEW WAN JIN
ID No	1211102837
Title of Thesis	GAIT RECOGNITION FROM SILHOUETTE IMAGES: A DEEP LEARNING APPROACH
Supervisor Name	PA PA MIN

REPORT ARRANGEMENT	√	**Comments** (if any differences)
1. Cover of The Final Report	√	
2. Title Page of the Final Report	√	
3 Copyright page of I Final Report	√	
4. Declaration Page of Final report	√	
5. Acknowledgement	√	
6. Table of Contents	√	
7. Abstract	√	
8. List of Tables	√	
9. List of Figures	√	
10. List of Symbols	√	
11. List of Appendices	√	
12. Chapter 1: Introduction – objectives, scope	√	
13. Chapter 2: Literature Review	√	
14. Chapter 3: Methodology	√	
15. Chapter 4: Experimental Design and Analysis	√	
16. Chapter 5: Conclusion	√	
20. References – APA style	√	
21. Appendices	√	
22. CD/ DVD and envelope as shown in Appendix K		
23. Attachment: FYP Meeting Logs (all) - 1 set	√	

FORMAT OF REPORT	√	**Comments**
1. Page Numbering	√	
2. Font and Type Face	√	
3. Font Cover	√	
4. Tables and Figures	√	
5. Spine Format		
6. Comb Bind (For evaluation)		
7. Permanent Bind (After approval)		
8. Colour of the Front Cover		
9. Number of words > 10000 (Main content only)	√	

Checked by

12/06/2025

Student's Signature & Date